LOGTEL
It's more than knowledge

IOT (and M2M and WoT)

Day 1 - Foundations | 2026

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The course: four days, one story

LOGTEL
It's more than knowledge

Day 1
Foundations

- Networking: OSI and TCP/IP
- Wi-Fi
- Wireshark: seeing packets
- Hardware interfaces
- What IoT is - and why now
- Bluetooth, NFC, RFID, UWB

TODAY

Day 2
Technologies

- Architecture and cloud
- LPWAN, cellular, satellite
- LoRa and LoRaWAN
- Zigbee, Thread, Matter
- MQTT and brokers
- Grafana dashboards
- MakeCode warm-up

Day 3
Cellular IoT and security

- Cellular IoT: NB-IoT, LTE-M
- D2D and V2X
- Device management, OTA updates
- End-to-end IoT security
- Cellular modem practice

Gilad

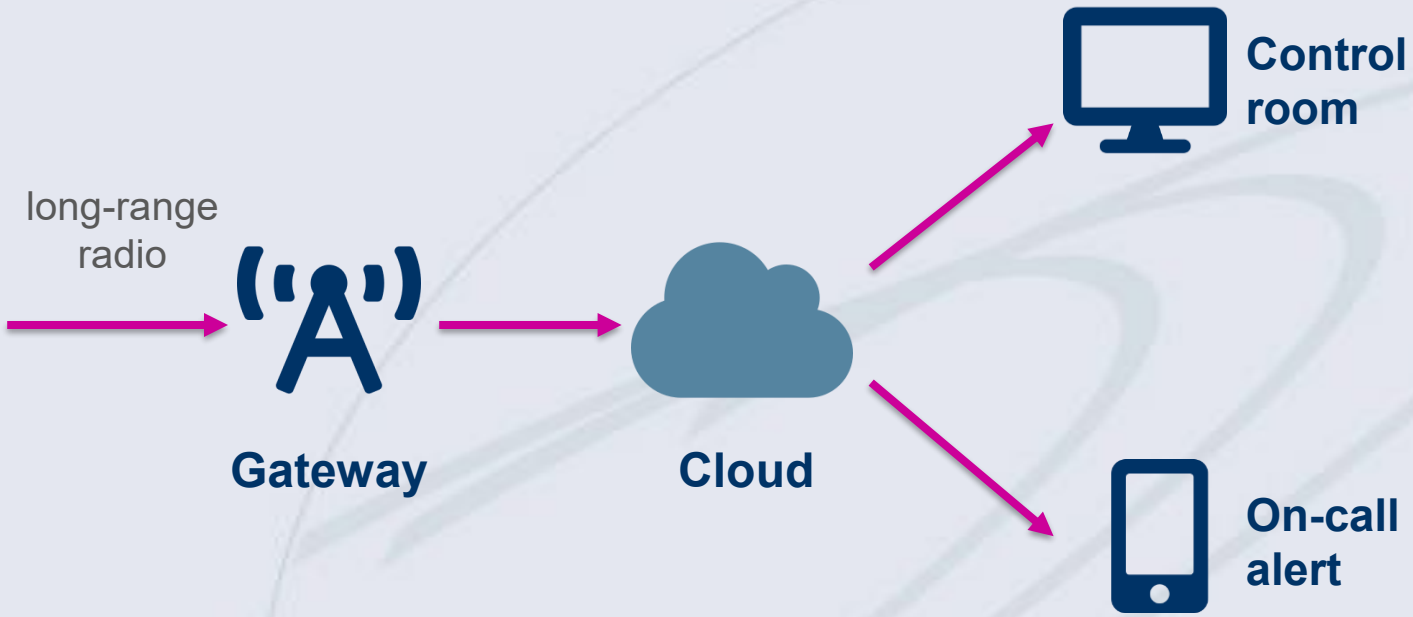
Day 4
Hands-on lab

- AWS IoT Core
- Certificates and mTLS
- micro:bit to the cloud
- MQTT + Wireshark + Grafana
- Cloud-to-device control
- Troubleshooting challenges

Our running scenario: Remote Site Watch

Unmanned remote site

- Temperature
- Door opened
- Vibration / tamper
- Water level
- Battery level



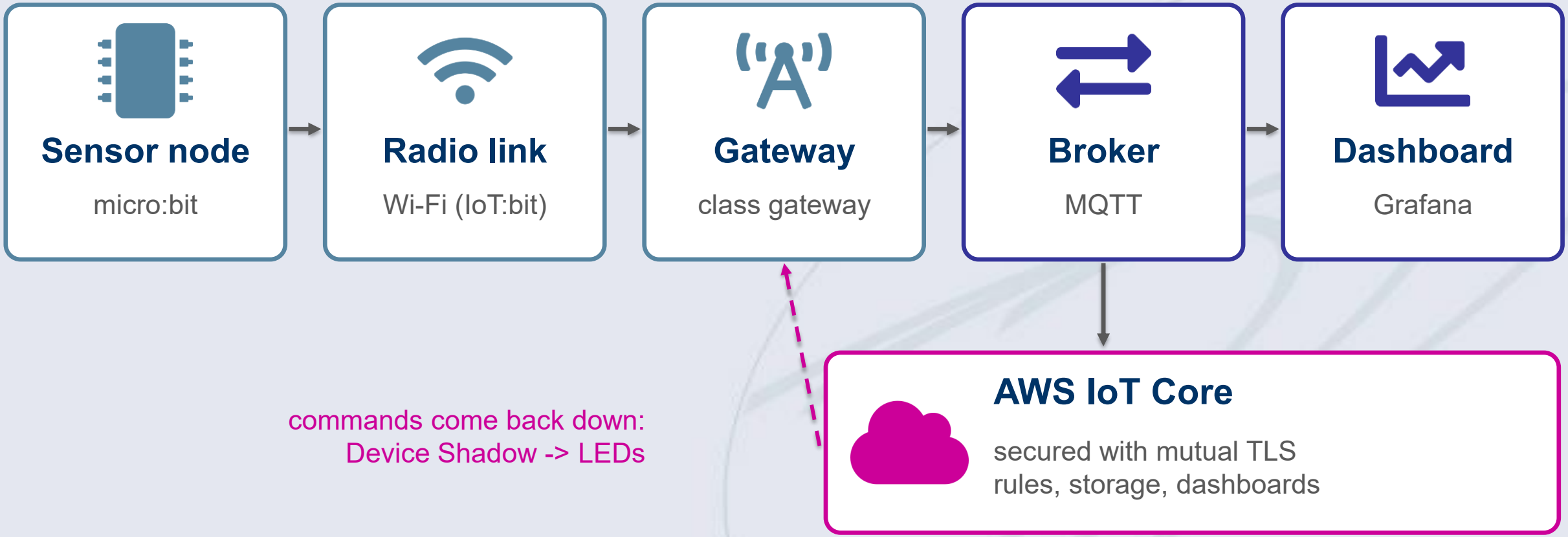
Day 1: the parts - sensors, interfaces, radios, networks

Day 2: the system - LoRaWAN, MQTT, dashboards

Day 3 (Gilad): cellular links, device management, end-to-end security

Day 4: we build a mini version - micro:bit to AWS IoT

The pipeline we build - one step per day



Day 1 - understand each part Day 2 - messaging, dashboards Day 4 - build it end to end

Day 1 - Foundations: today's plan

Morning

1. **Networking basics**
   OSI and TCP/IP - layers 4, 5, 7
2. **Wi-Fi**
   layers 1-2 on the air, joining a network
3. **Wireshark**
   live demo + 5-minute try
4. **Hardware interfaces**
   GPIO, UART, I2C, SPI

After lunch

5. **Introduction to IoT**
   uses, threats, rules
6. **More short-range radios**
   Bluetooth, NFC, RFID, UWB
7. **Summary and quiz**
   what to remember

DATA NETWORKING

Why layers? Think of sending a letter

The post office		The network	
 The letter	⟷	Application data	temp = 23.5
 Name on the envelope	⟷	Transport: port	MQTT, port 8883
 Street address	⟷	Network: IP address	18.192.x.x
 Mail-bag, next truck	⟷	Link: Wi-Fi frame	next hop only
 Road or air	⟷	Physical: radio, cable	2.4 GHz

Each layer adds its own envelope - the receiver opens them in reverse order.

OSI 7 Layer Model

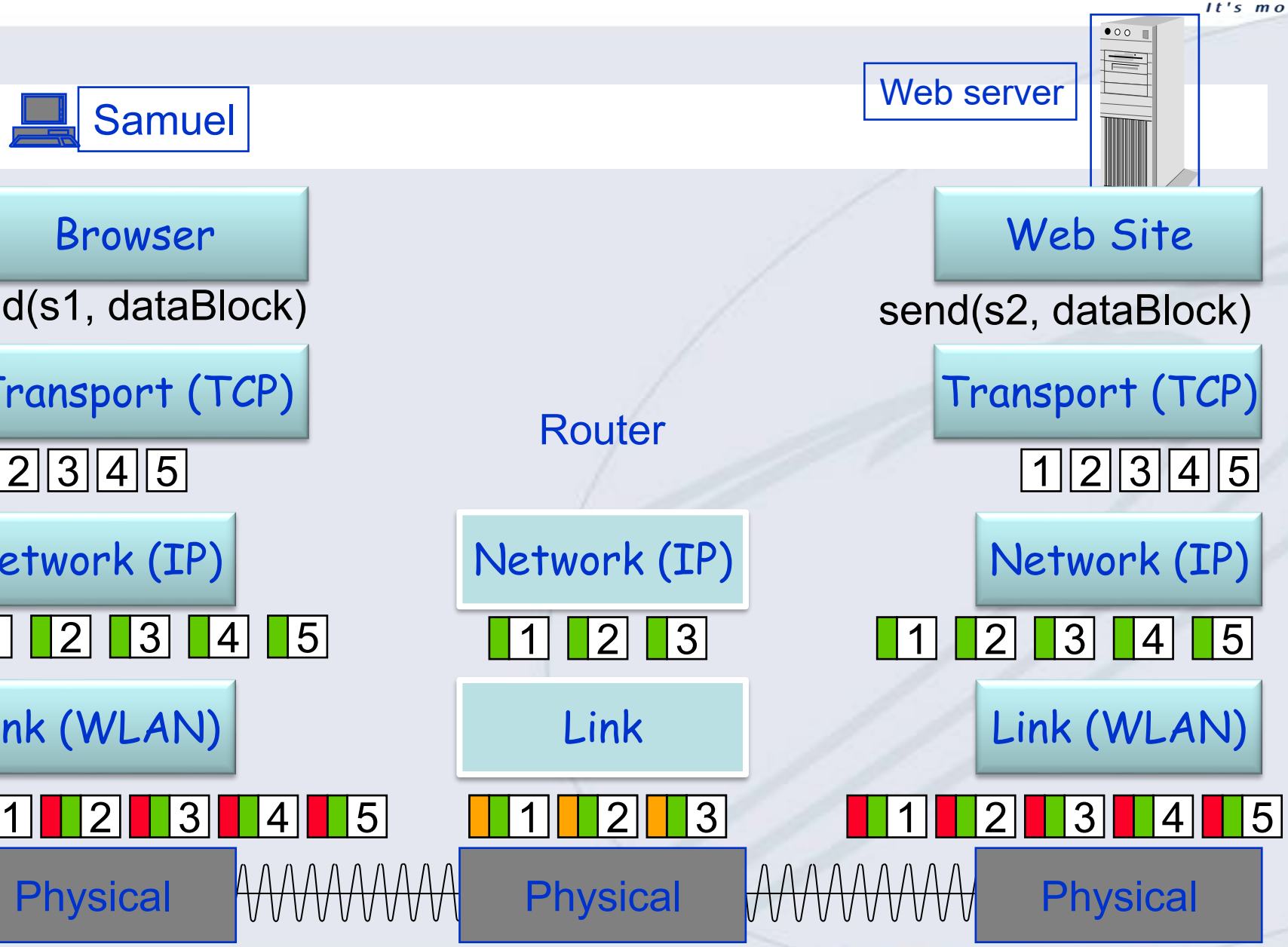
- OSI - Open Systems Interconnection (Basic Reference Model)
- Each level is an independent set of protocols
- Each level can be change seamlessly



The 7 OSI layers in plain words

#	Layer	Its job	IoT example
7	Application	The service the device wants	MQTT, HTTP, CoAP
6	Presentation	Format, compress, encrypt	JSON, CBOR, TLS
5	Session	Open, keep and close a dialogue	MQTT session, keep-alive
4	Transport	Right application; reliable or fast	TCP, UDP, ports
3	Network	Addresses and routes	IPv4, IPv6
2	Data link	One hop; who may talk now	Wi-Fi, BLE, LoRaWAN MAC
1	Physical	Bits as signals	radio, cable, fiber

Our focus: layers 7, 5 and 4 - and just enough of 1-3 to follow the data.


Samuel
Web server
Browser
Web Site
read(s1, dataBlock)
send(s2, dataBlock)
Transport (TCP)
Transport (TCP)
Router
1 2 3 4 5
1 2 3 4 5
Network (IP)
Network (IP)
Network (IP)
1 2 3 4 5
1 2 3
1 2 3 4 5
Link (WLAN)
Link
Link (WLAN)
1 2 3 4 5
1 2 3
1 2 3 4 5
Physical
Physical
Physical

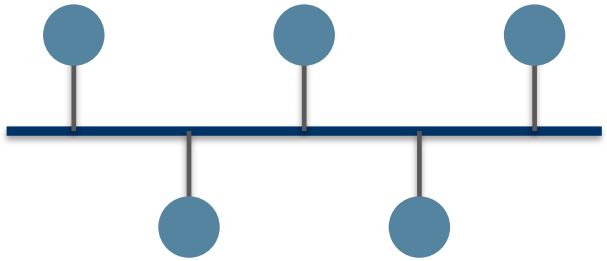
Where IoT protocols live

	Classic Internet	IP-based IoT	Non-IP IoT
Application	HTTP	MQTT, CoAP	raw bytes
Security	TLS	TLS, DTLS	link encryption
Transport	TCP	TCP or UDP	-
Network	IPv4 / IPv6	IPv6, 6LoWPAN	no IP
Link / radio	Ethernet, Wi-Fi	Thread, NB-IoT	LoRaWAN, BLE, Zigbee

Many IoT devices never speak IP - a gateway translates them into the Internet.

Network topologies: the shape of a network

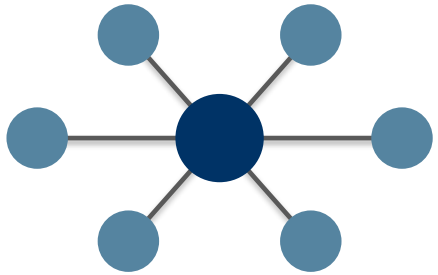
Bus



one shared line

IoT: I2C, a car's CAN bus

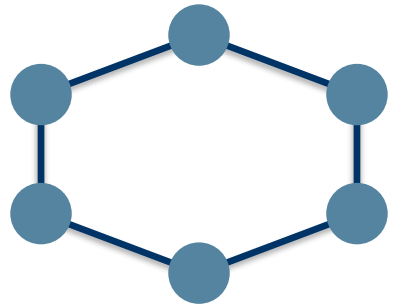
Star



everything goes through the centre

IoT: Wi-Fi, Bluetooth, cellular

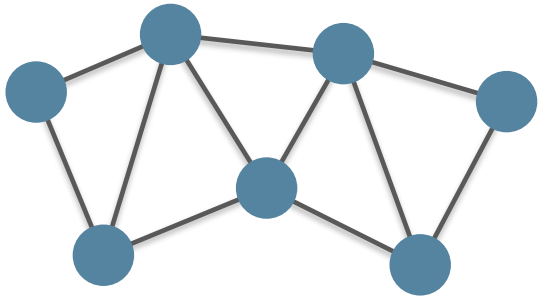
Ring



each node passes data on

IoT: industrial Ethernet rings

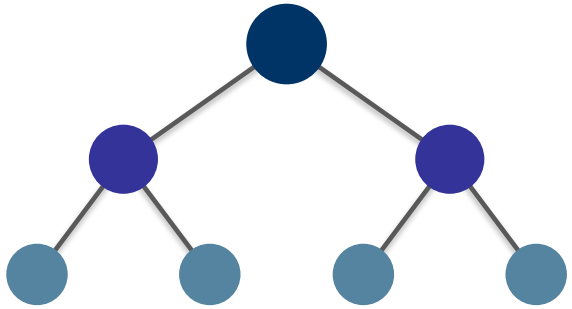
Mesh



nodes relay for each other

IoT: Zigbee, Thread, Bluetooth Mesh

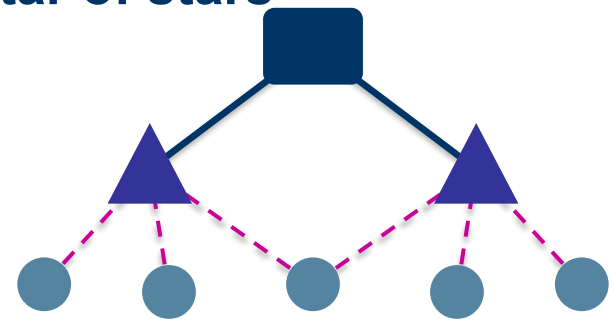
Tree



stars connected in levels

IoT: building and campus networks

Star of stars



devices -> any gateway -> one server

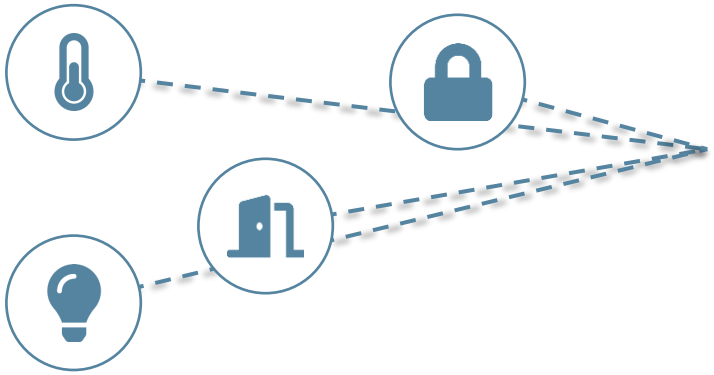
IoT: LoRaWAN (Day 2)

Gateway: joining different networks

Often used to connect a LAN with a WAN.
Gateways join two or more **different** networks together.

Local network

Zigbee, Bluetooth, LoRa ...



Gateway

translates
filters, secures

Wide area: Internet, cloud

IP, TCP, MQTT, HTTPS

apps, dashboards

Home

a Zigbee or Matter hub

In the field

a LoRaWAN gateway (Day 2)

Our lab

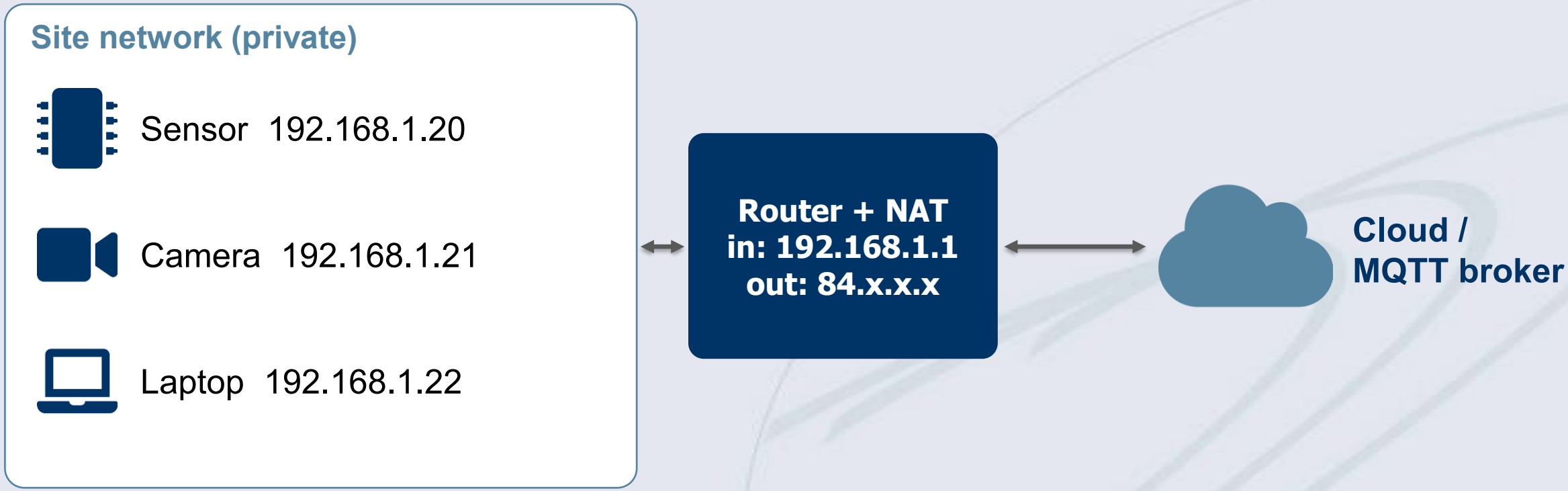
the class gateway: MQTT in, AWS IoT Core out (Day 4)

IP Address

- Unique addresses in the world
- An IP address is 32 bits, noted in dotted decimal notation:
  192.78.32.2

- Host and Prefix Part
  - An IP address has a prefix and a host part:
    - **prefix:host**
  - Prefix identifies a subnetwork
    - used for locating a subnetwork - routing
  - Prefix is usually identified in a host using a "subnet mask"

Private addresses and NAT



- Private addresses are not reachable from the Internet
- So the device dials OUT and keeps the connection open - commands come back on it

IPv6 and IoT

4.3 billion

IPv4 addresses in total

21+ billion

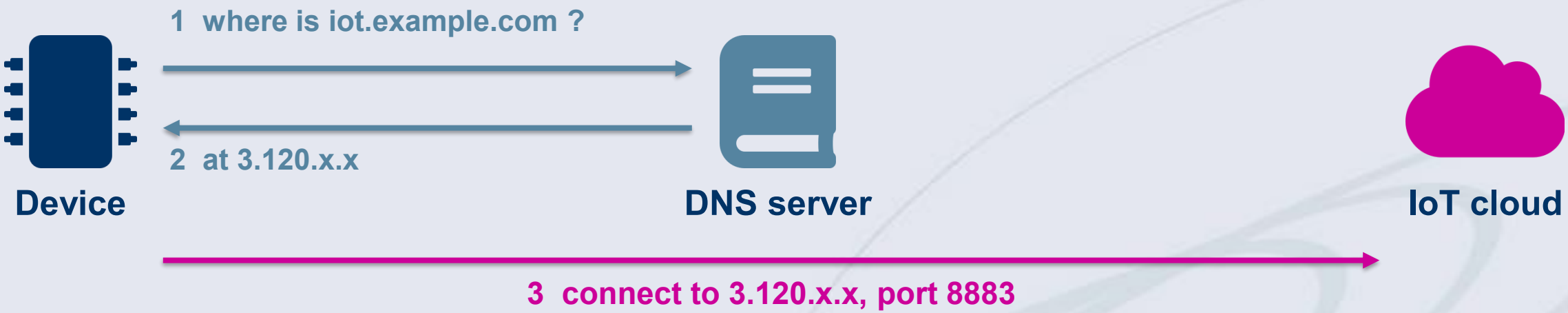
connected IoT devices (2025)

3.4 x 10^38

IPv6 addresses

- IPv4 survives thanks to NAT - fine for devices that only dial out
- IPv6: every device can have its own global address
- **6LoWPAN** squeezes IPv6 into tiny radio frames; **Thread** (used by Matter) is IPv6 end to end
- Cellular IoT and the big clouds run both (dual-stack)

DNS: the Internet's phone book



- Names for people and devices, numbers for the network - DNS translates
- The first thing a device does on a new network - and a classic failure point
- Never hard-code a cloud IP address in a device: names stay, addresses change
- Wireshark filter: **dns**

DHCP

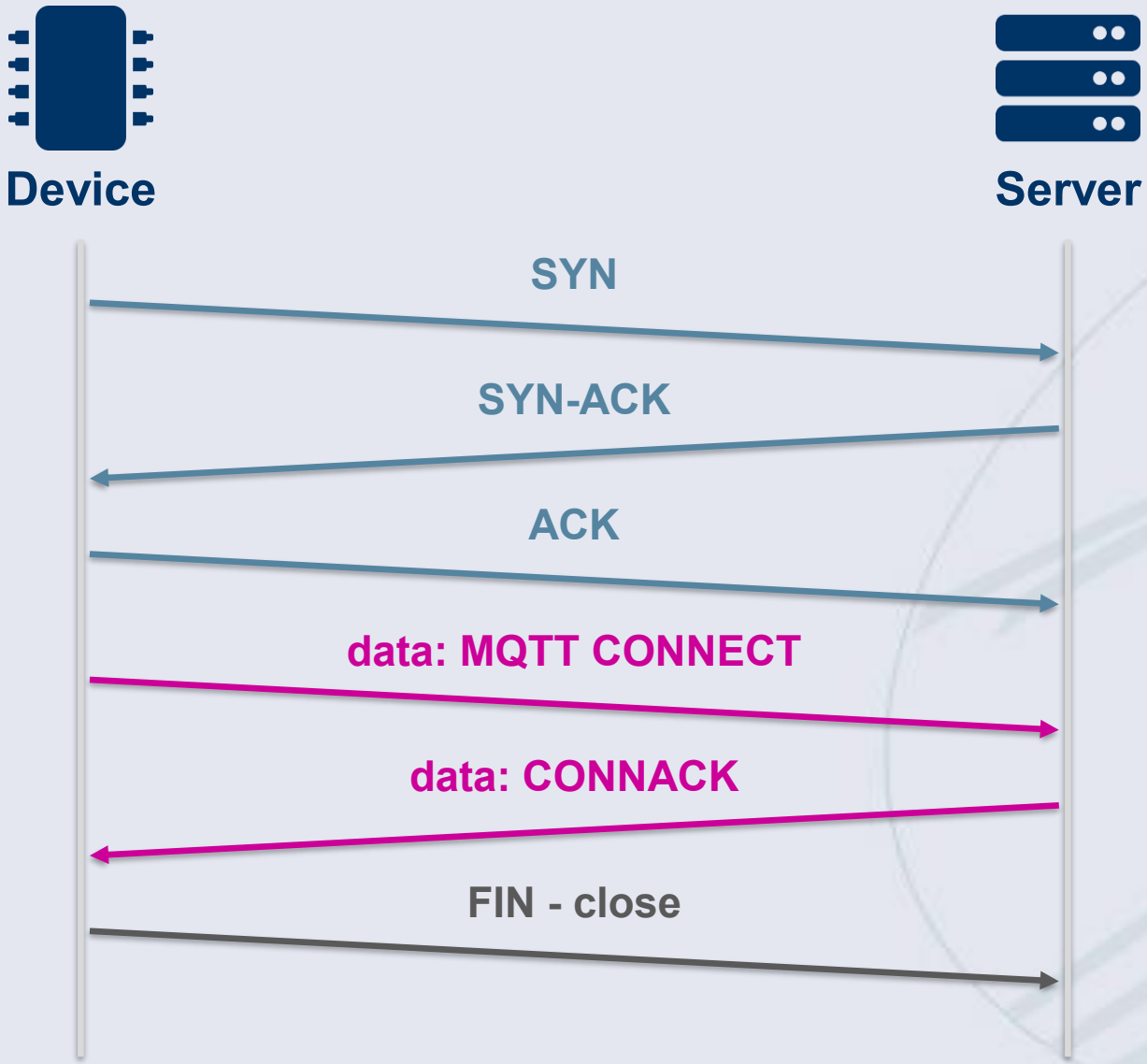
- Dynamic Host Configuration Protocol (DHCP) is designed, to dynamically configure TCP/IP hosts in a centralized manner from DHCP server.
- DHCP server maintains a collection of configuration parameters, such as IP addresses, subnet mask, default gateway IP address, to make a configured host work in the network.
- A DHCP client queries the server for the configuration parameters.
- The DHCP server returns configuration parameters to the client.

Layer 4: ports - the apartment number

Port	Protocol	What IoT uses it for
53	DNS (UDP)	find the server's address
80 / 443	HTTP / HTTPS (TCP)	web and cloud APIs
1883	MQTT (TCP)	message broker, no encryption
8883	MQTT over TLS (TCP)	secure MQTT - AWS IoT Core
5683	CoAP (UDP)	tiny battery devices
1700	LoRa packet forwarder (UDP)	LoRaWAN gateway to network server

IP address = which building. Port = which apartment (which program) inside it.

TCP: a handshake before talking



- 3 messages before any data
- Every byte is numbered and acknowledged - losses are re-sent
- Costly on a battery: MQTT keeps ONE connection open for hours
- Wireshark: **tcp.flags.syn == 1**

TCP or UDP? Reliable vs fast

	TCP	UDP
Connection	handshake first	none - just send
Reliability	acknowledged, re-sent, in order	best effort
Overhead	20+ byte header, ACKs	8 byte header
IoT examples	MQTT, HTTPS, AWS IoT	DNS, CoAP, LoRa forwarder
Analogy	registered mail	a postcard

Modern twist: QUIC (the base of HTTP/3) adds reliability on top of UDP.

Layer 5 in IoT: sessions that stay alive



- A session remembers who you are and what you subscribed to
- **Keep-alive**: small pings, so both sides notice a dead link
- **Last Will**: the broker announces "site-7 offline" for you
- Reconnect and receive what you missed - all of this is MQTT, Day 2

Layer 6: one reading, three formats

Format	What is sent	Size
JSON (text)	{"temp":23.5,"bat":87}	22 bytes
CBOR (binary JSON)	A2 64 74 65 6D 70 F9 4D E0 ...	15 bytes
Packed binary	00 EB 57	3 bytes

- JSON is easy to read - but every character costs
- A long-range LoRaWAN message may carry only about 50 bytes
- So devices send compact bytes; a **decoder** turns them back into JSON (Day 2)

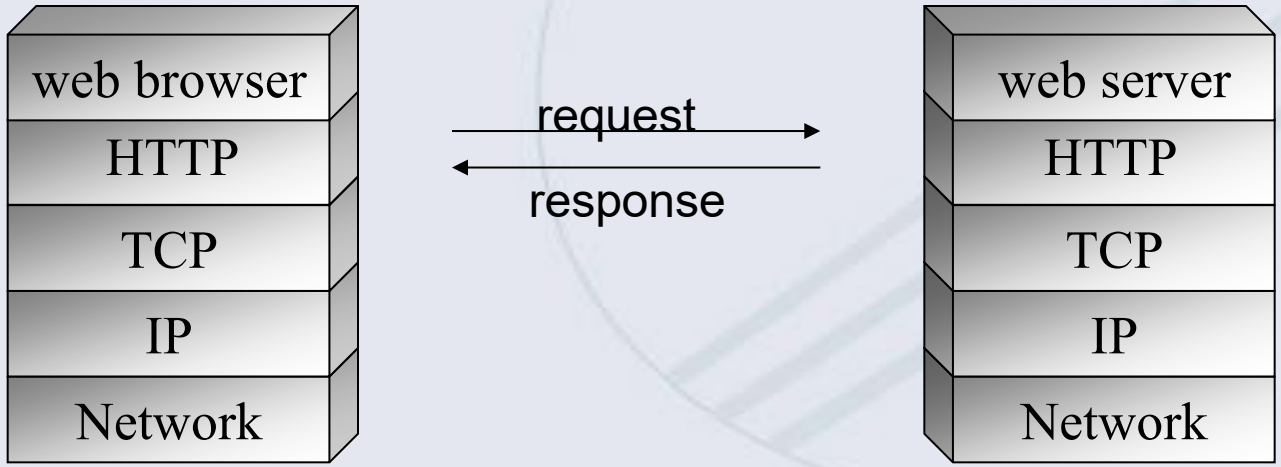
TLS: the padlock



- **Privacy** (encryption), **integrity** (no tampering), **identity** (the real server)
- HTTPS = HTTP + TLS; MQTT + TLS = port 8883
- Normal TLS proves only the server **- mutual TLS** (Day 4) makes the device prove itself too

HyperText Transfer Protocol (HTTP)

- Application layer protocol
- Distributes information in the WWW
- Based on the client/server architecture
  - HTTP client (web browser): sends a request to a server for a file
  - HTTP server (web server): well-known port number 80, responds with the requested file if it is available
  - A single TCP connection is used



Layer 7: IoT application protocols

Protocol	Pattern	Best for
HTTP / REST	request - response	cloud APIs, occasional uploads
MQTT	publish - subscribe (broker)	telemetry and commands
CoAP	request - response over UDP	tiny battery devices
LwM2M	device management	firmware, configuration
AMQP	message queues	enterprise back-ends

HTTP: the device asks, the server answers.
MQTT: everyone talks to a broker - the star of Day 2.

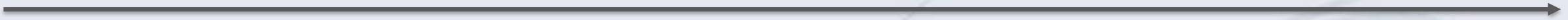
WI-FI



Layers 1 and 2 on the air - and what happens when a device joins

Radio basics for non-engineers

13.56 MHz	sub-GHz	2.4 GHz	5 / 6 GHz	6-8 GHz
NFC	LoRa, HaLow, RFID	Wi-Fi, Bluetooth, Zigbee	Wi-Fi 5-7	UWB



lower frequency — higher frequency

Lower frequency

- travels farther
- passes walls
- less data

Higher frequency

- shorter range
- easily blocked
- more data

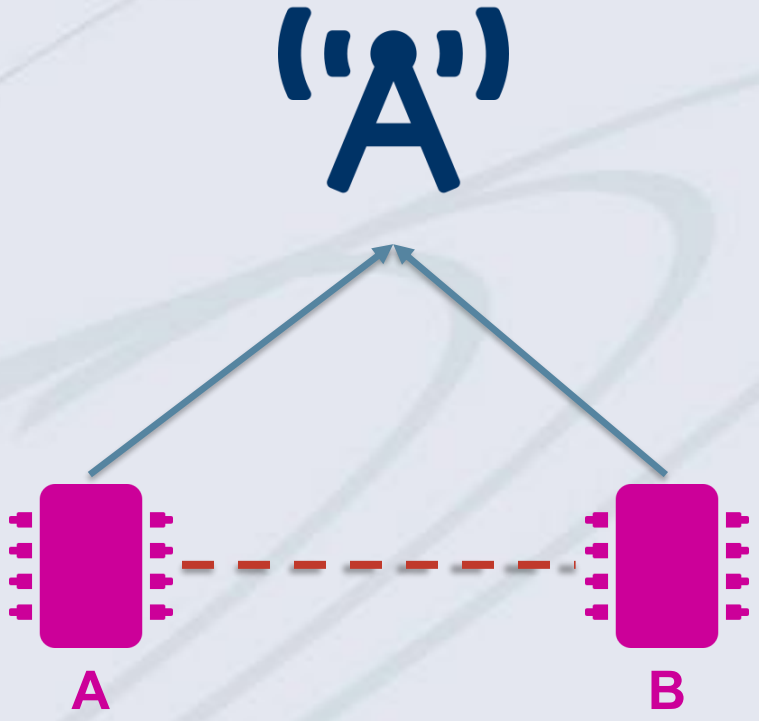
Pick two

- long range
- high data rate
- low power

Licensed bands (cellular): paid and protected. Unlicensed (ISM): free for all - so crowded, with power and duty-cycle rules.

Sharing the air

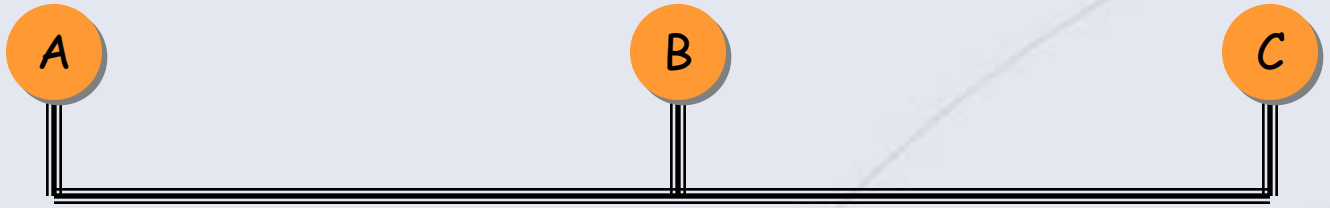
- On a cable you can listen while you talk - on radio you cannot
- Two devices may both reach the access point but not hear each other
- So Wi-Fi listens first, waits, and can reserve the air (RTS / CTS)
- More devices on a channel = more waiting
- LPWAN does the opposite: send rarely, accept collisions (Day 2)



the 'hidden node' problem

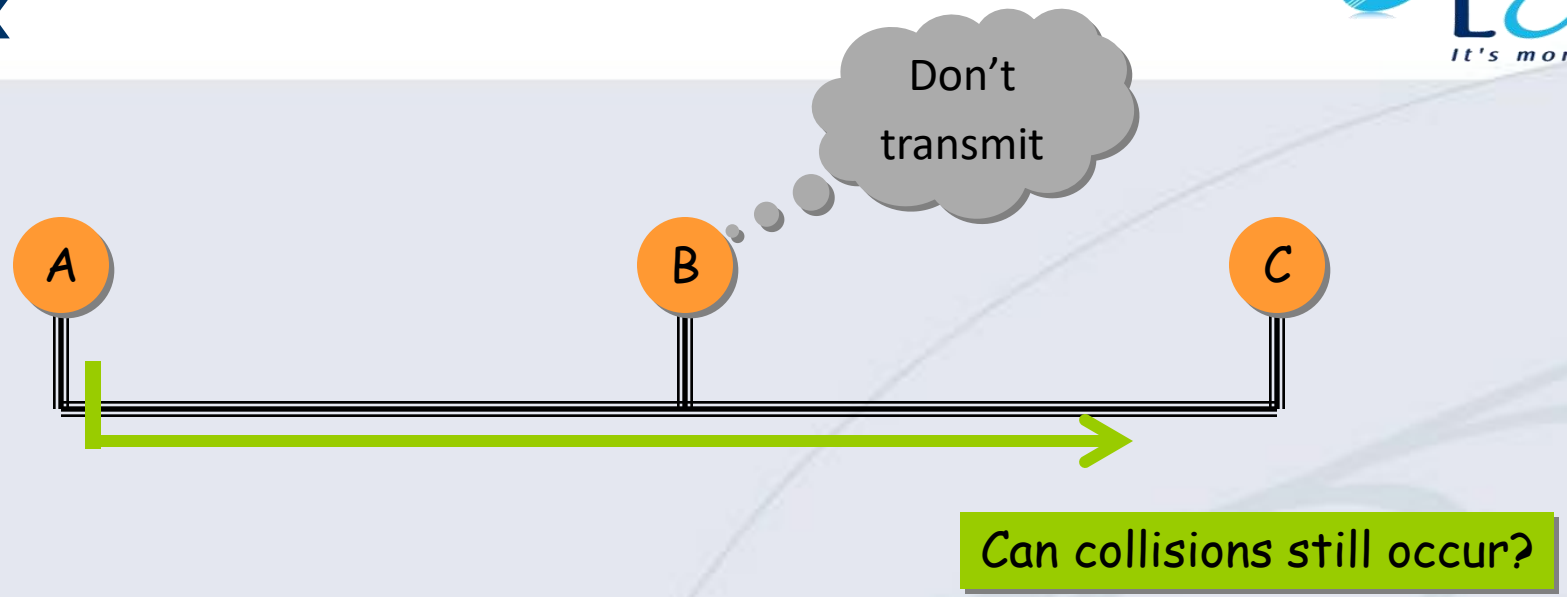
The Channel Access Problem

- Multiple nodes share a channel



- Pairwise communication desired
  - Simultaneous communication not possible
- MAC Protocols
  - Suggests a scheme to schedule communication
    - Maximize number of communications
    - Ensure fairness among all transmitters

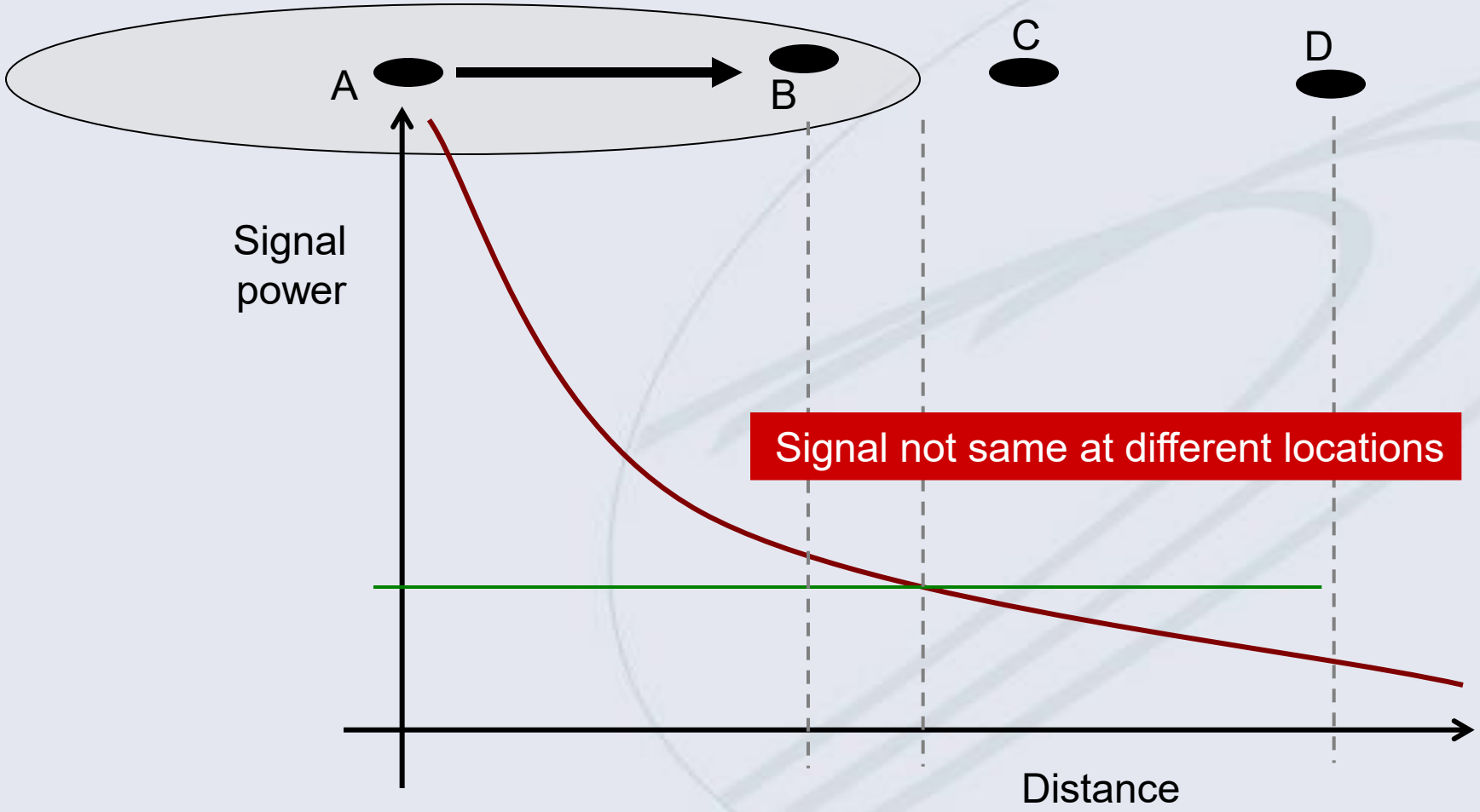
The Simple Fix



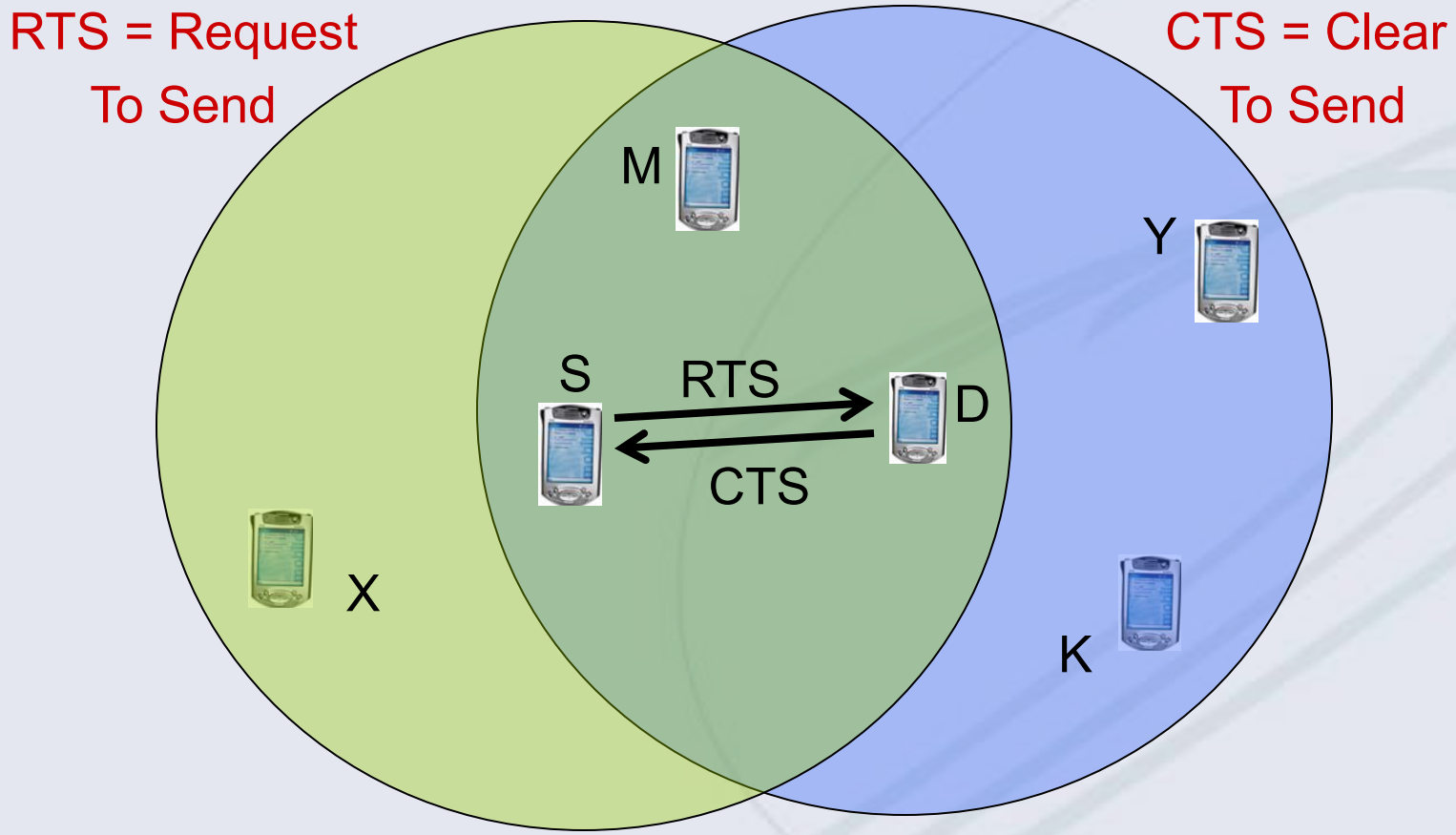
- Transmit and pray
  - Plenty of collisions --> poor throughput at high load
- Listen before you talk
  - Carrier sense multiple access (CSMA)
  - Defer transmission when signal on channel

Wireless Media Disperse Energy

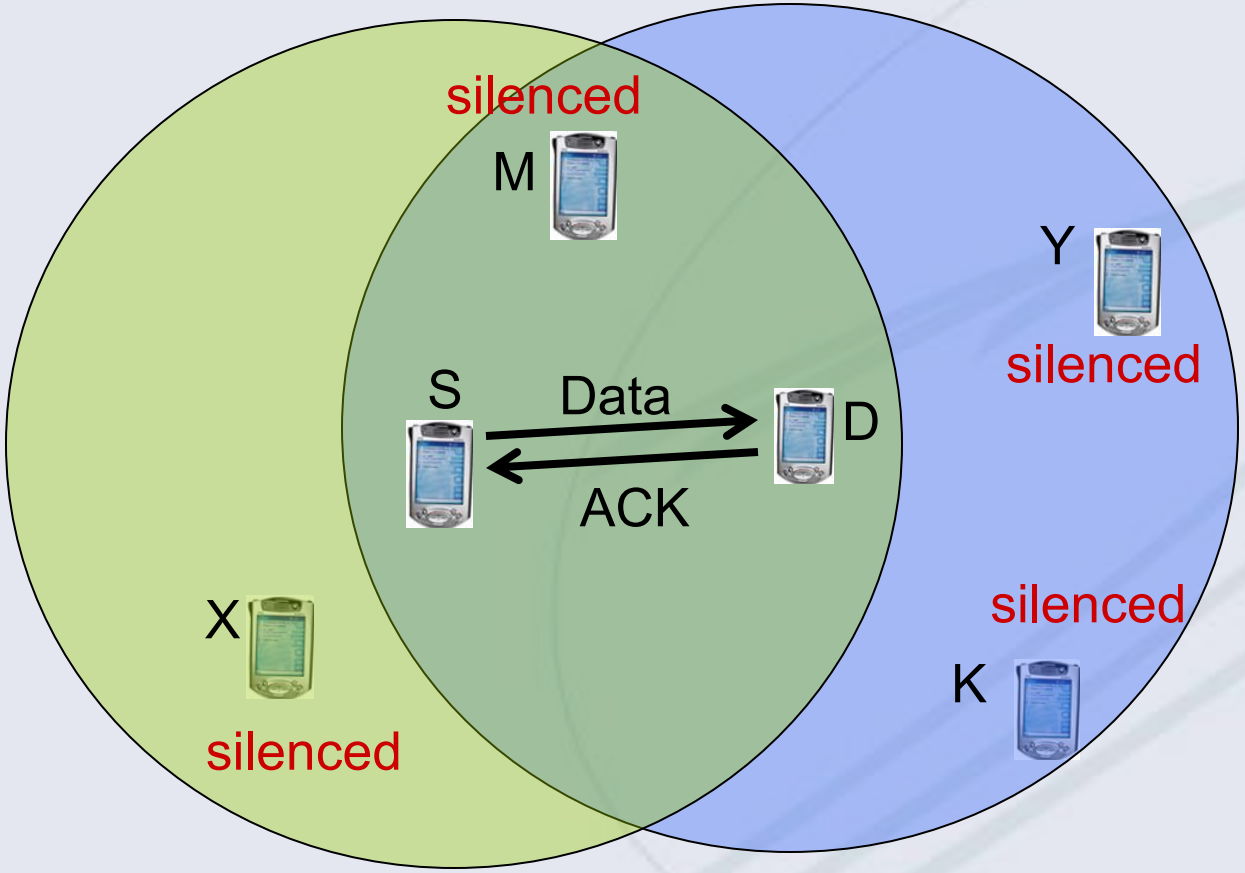
A cannot send and listen in parallel



IEEE 802.11



IEEE 802.11



Wi-Fi generations

Name	IEEE	Certified	Max rate (theory)	Bands
Wi-Fi 4	802.11n	2009	600 Mbit/s	2.4 + 5 GHz
Wi-Fi 5	802.11ac	2013	6.9 Gbit/s	5 GHz
Wi-Fi 6	802.11ax	2019	9.6 Gbit/s	2.4 + 5 GHz
Wi-Fi 6E	802.11ax	2021	9.6 Gbit/s	+ 6 GHz
Wi-Fi 7	802.11be	2024	~23 Gbit/s	2.4 + 5 + 6 GHz
Wi-Fi 8	802.11bn	~2028	focus: reliability	1 - 7.25 GHz

- Wi-Fi 7 is now mainstream: over half of new access points sold in Q2 2026
- Real devices reach a few Gbit/s - the table shows theoretical peaks

Wi-Fi for IoT: what changed recently



Target Wake Time

Wi-Fi 6 sensors agree a schedule and sleep for hours



Multi-Link (Wi-Fi 7)

several bands at once: steadier links for industry



Wi-Fi sensing (802.11bf)

2025 standard: Wi-Fi itself detects presence and motion



6 GHz in Israel

lower 6 GHz band (5925-6425 MHz) open for Wi-Fi 6E / 7

Wi-Fi is the #1 IoT connection type: 32% of all IoT devices (2025).

Wi-Fi HaLow (802.11ah): Wi-Fi for long range

~1 km	sub-GHz	8,191	16 km
typical range	passes walls	devices per access point	record video call (2024)

- Normal Wi-Fi security and IP - but below 1 GHz and with low power
- Commercial deployments started in 2025; modules in mass production in 2026
- Sits between Wi-Fi and LoRaWAN: more data than LoRa, more range than Wi-Fi
- Frequencies differ by country - check local rules before buying

What happens when a device joins Wi-Fi

1	**Scan**	find the access point
2	**Associate**	join it
3	**WPA2 / WPA3**	agree on encryption keys
4	**DHCP**	get an IP address
5	**DNS**	find the server
6	**TCP + TLS**	secure connection
7	**MQTT**	start sending data

- Seven steps before the first reading leaves the device
- Each step can fail - and each failure looks different in Wireshark
- A sleeping device repeats much of this when it wakes: Wi-Fi is power-hungry

Wireshark: every layer of every packet

Capturing from Wi-Fi

```
dns || tcp.port == 1883
```

No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000	192.168.1.23	192.168.1.1	DNS	79	Standard query A gateway.example
2	0.011824	192.168.1.1	192.168.1.23	DNS	95	Standard query response A 203.0.113.10
3	0.014031	192.168.1.23	203.0.113.10	TCP	66	49152 → 1883 [SYN]
4	0.061217	203.0.113.10	192.168.1.23	TCP	66	1883 → 49152 [SYN, ACK]
5	0.061490	192.168.1.23	203.0.113.10	TCP	54	49152 → 1883 [ACK]
6	0.062113	192.168.1.23	203.0.113.10	MQTT	99	Connect Command
7	0.108754	203.0.113.10	192.168.1.23	MQTT	58	Connect Ack
8	**0.110386**	**192.168.1.23**	**203.0.113.10**	**MQTT**	**101**	**Publish Message [lab/team03/telemetry]**

L1 > Frame 8: 101 bytes on wire (808 bits), 101 bytes captured

L2 > Ethernet II, Src: 02:00:00:aa:00:23, Dst: 02:00:00:bb:00:01

L3 > Internet Protocol Version 4, Src: 192.168.1.23, Dst: 203.0.113.10

L4 > Transmission Control Protocol, Src Port: 49152, Dst Port: 1883, Len: 47

L7 v MQ Telemetry Transport Protocol, Publish Message

Topic: lab/team03/telemetry

Message: {"temp":23,"light":120}

```
0000   02 00 00 00 bb 01 02 00  00 00 aa 23 08 00 45 00   ...........#..E.
0010   00 57 1c 46 40 00 40 06  20 91 c0 a8 01 17 cb 00   .W.F@.@. .......
0020   71 0a c0 00 07 5b 1a 2b  3c 4d 5e 6f 70 81 50 18   q....[.+<M^op.P.
0030   fa f0 e4 2b 00 00 30 2d  00 14 6c 61 62 2f 74 65   ...+..0-..lab/te
0040   61 6d 30 33 2f 74 65 6c  65 6d 65 74 72 79 7b 22   am03/telemetry{"
0050   74 65 6d 70 22 3a 32 33  2c 22 6c 69 67 68 74 22   temp":23,"light"
0060   3a 31 32 30 7d                                    :120}
```

- L2 Ethernet / Wi-Fi
- L3 IP
- L4 TCP
- L7 MQTT

Wireshark: cheat sheet and a 5-minute try

Display filter	Shows
dns	name look-ups
tcp.flags.syn == 1	new connections
tls.handshake	TLS handshakes
mqtt	MQTT (Days 2 and 4)
ip.addr == 192.168.1.20	one device

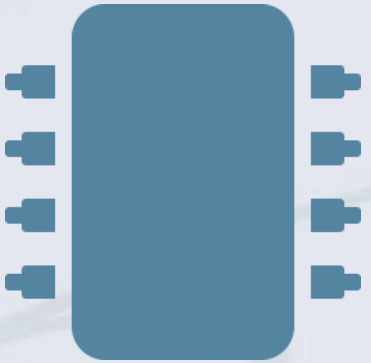
Three panes: packet list, packet details (the layers), raw bytes.

Try it - 5 minutes

- Capture on Wi-Fi
- Open a web site
- Filter **dns** - find question and answer
- Filter **tcp.flags.syn == 1**
- Name the layers of one packet

Capture only where you are authorized.

LOGTEL
It's more than knowledge



HARDWARE INTERFACES

How sensors talk to the controller: GPIO, UART, I2C, SPI

Inside an IoT device



Sensors

temperature, motion, light



Radio

BLE, Wi-Fi, LoRa, cellular

Microcontroller (MCU)



Actuators

LED, relay, motor, display



Power

battery, harvesting

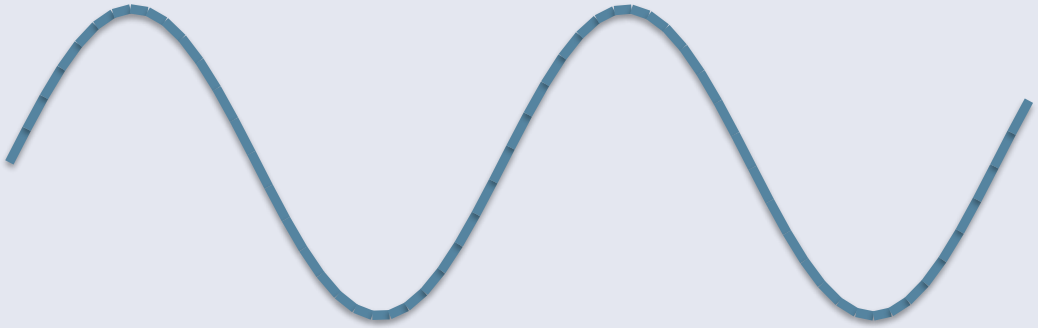
Today's question: what are the wires between these chips?

A microcontroller is not a small PC

	micro:bit V2	Raspberry Pi 5	Laptop
Processor	64 MHz, 1 core	2.4 GHz, 4 cores	3-5 GHz, many cores
Memory	128 KB RAM	2-16 GB RAM	16-64 GB RAM
Software	one program, no OS	Linux	Windows / macOS
Power	milliwatts	watts	tens of watts
IoT role	sensor node	gateway, edge	management

Kilobytes of memory and milliwatts of power: that is why IoT protocols must be small, simple and sleepy.

Analog vs digital - and the ADC



Analog: any value, continuously



Digital: only 0 or 1



- The ADC (analog-to-digital converter) turns a voltage into a number
- micro:bit: 0-3.3 V reads as 0-1023 - half the voltage is about 512

GPIO: one pin, one bit

Input

- button pressed?
- door open?
- motion?

Output

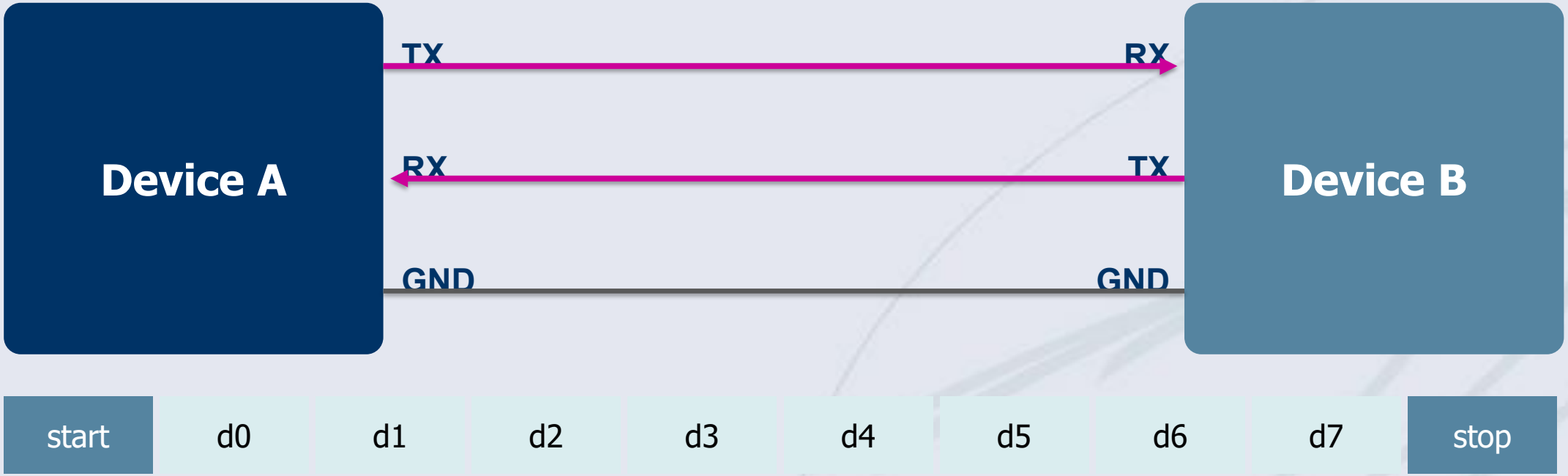
- LED on / off
- buzzer
- relay: switch a pump

PWM / analog

- dim an LED
- drive a servo
- read a light sensor

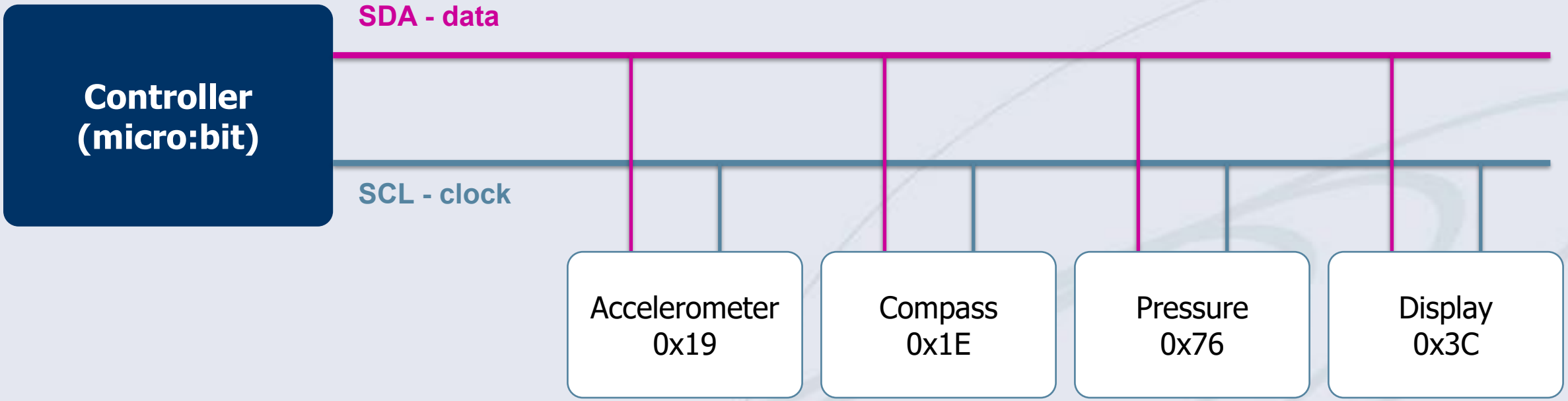
- **General Purpose Input/Output**: a pin read or driven high (3.3 V) or low (0 V)
- No protocol, no address - one signal per pin
- micro:bit: big rings 0, 1, 2 - and buttons A/B are GPIO inputs

UART: two wires and an agreed speed



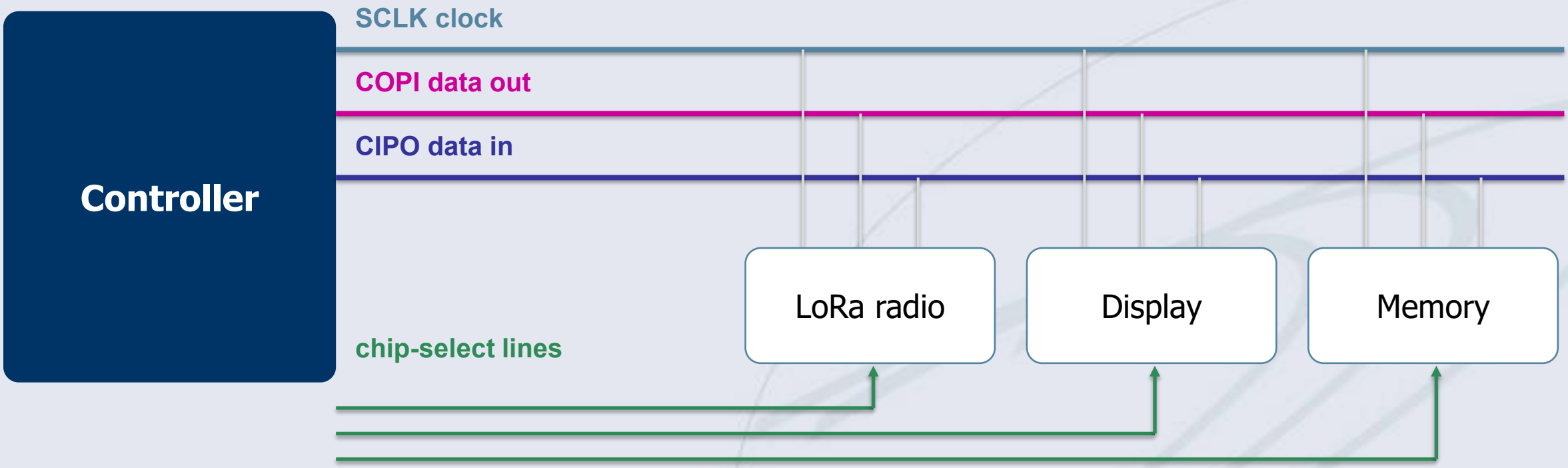
- No clock wire - both sides agree on the speed (baud), e.g. 115200
- GPS modules, modems (AT commands), debug consoles
- Our lab: the micro:bit drives its Wi-Fi chip over UART (AT commands)

I2C: two wires, many addressed devices



- The controller calls each device by its **address**
- 100 kbit/s to 3.4 Mbit/s; short, on-board wiring
- micro:bit V2: the motion sensor sits on an internal I2C bus

SPI: faster, one 'chip select' per device



- 4 wires + one chip select per device; MHz to tens of MHz
- Radios, displays, memory - every LoRa module is an SPI device

Choosing an interface

	Wires	Speed	Devices	Typical use
GPIO	1 per signal	-	1	buttons, LEDs, relays
UART	2 + ground	up to ~1 Mbit/s	2	GPS, modems, PC
I2C	2 + ground	up to 3.4 Mbit/s	many	sensors
SPI	4 + 1 each	tens of Mbit/s	several	radios, displays

RS-485 / Modbus

long cables in factories and utilities

CAN bus

vehicles, drones, robots

MIPI I3C

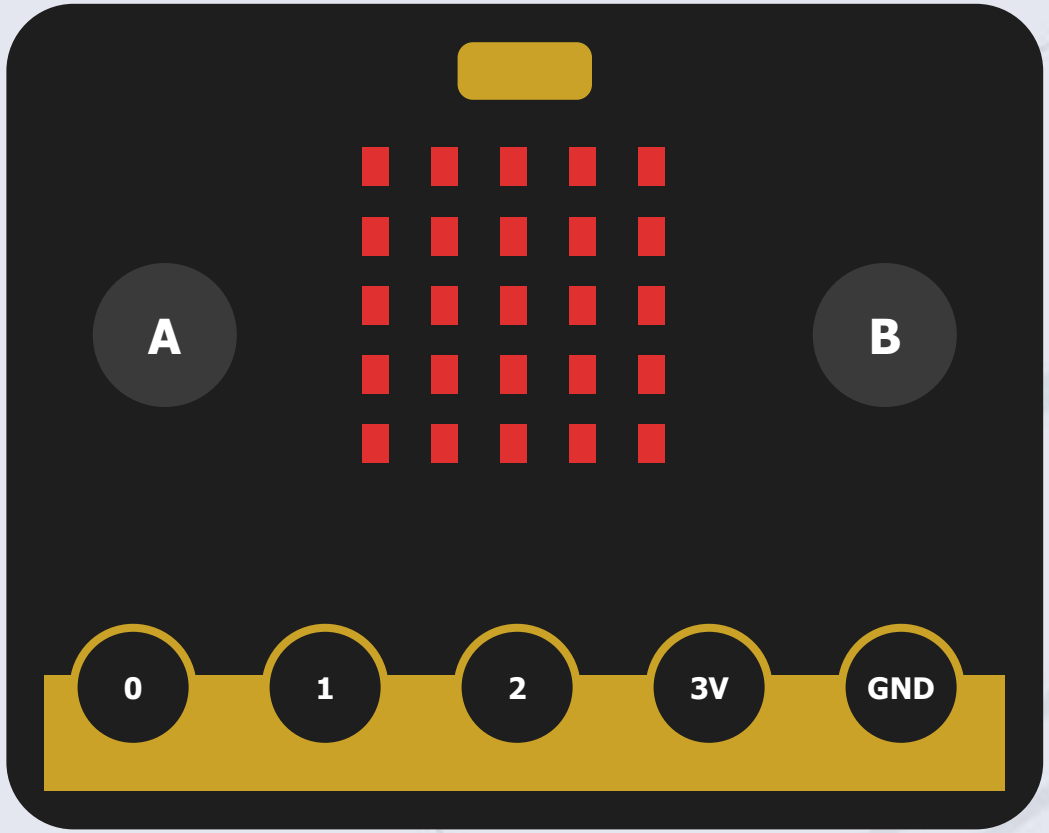
faster, lower-power successor of I2C

Our lab controller: BBC micro:bit V2

64 MHz processor
128 KB RAM

Bluetooth LE +
2.4 GHz radio

USB: power, flash,
serial to PC

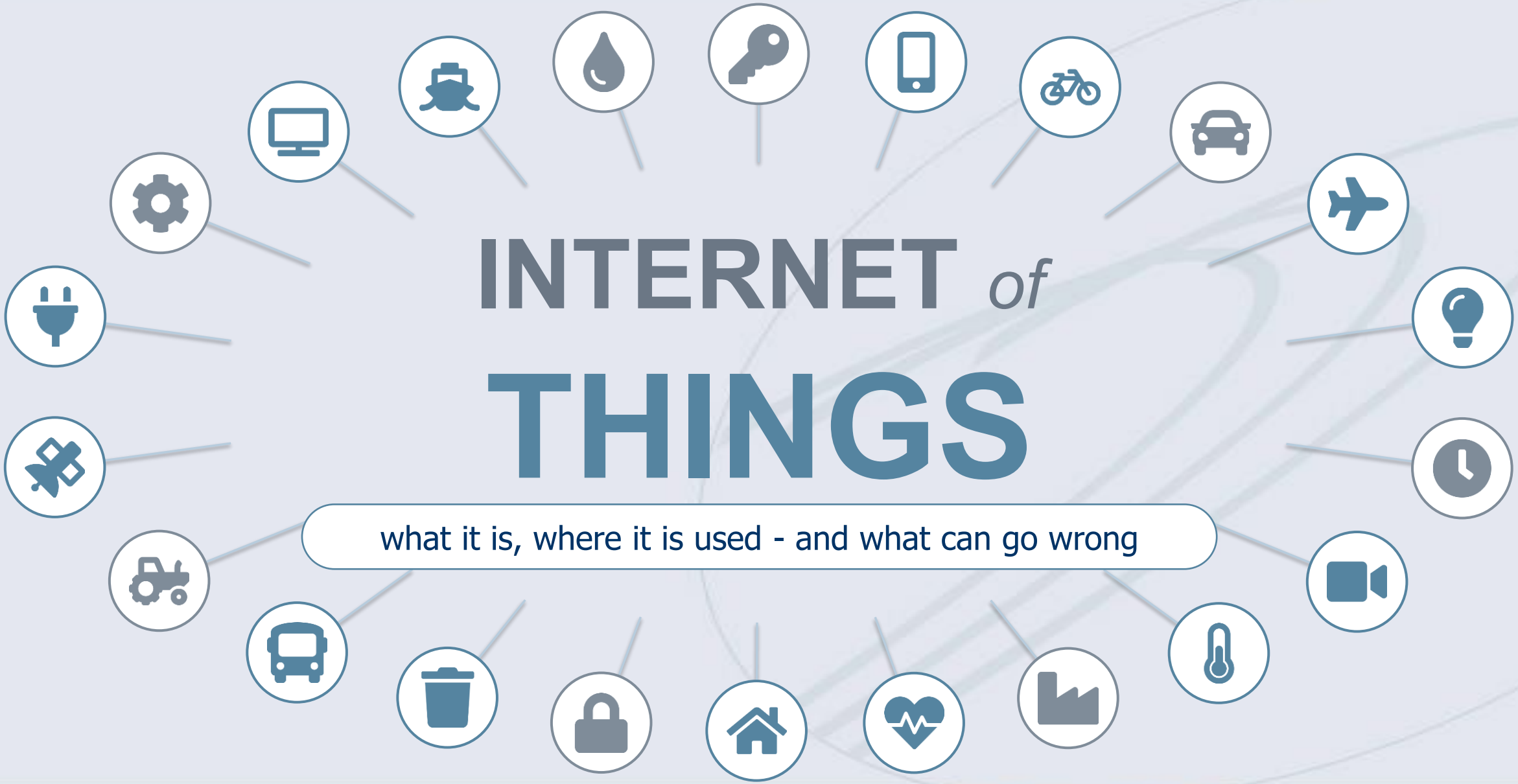


Motion sensor
(on I2C)

Temperature, light,
mic, speaker

Edge pins: GPIO,
I2C, SPI, UART

In the lab it sits on an ELECFREAKS IoT:bit board that adds Wi-Fi - the way a tiny sensor node reaches a network.



INTERNET *of* THINGS

what it is, where it is used - and what can go wrong

What is M2M / IoT / WoT?


RuinDig/Yuki Uchida, CC BY 4.0


Steffen Mokosch, CC BY-SA 4.0


Tony Webster, CC BY 2.0

M2M: machine to machine

One machine reports to one fixed destination. No human in the loop.

Today: EU eCall: after a crash the car dials 112 and sends its location (all new models since 2018)

IoT: Internet of Things

Many devices, one cloud platform, many apps reusing the data.

Today: 7 million+ public EV chargers worldwide (end of 2025), typically managed from cloud systems

WoT: Web of Things

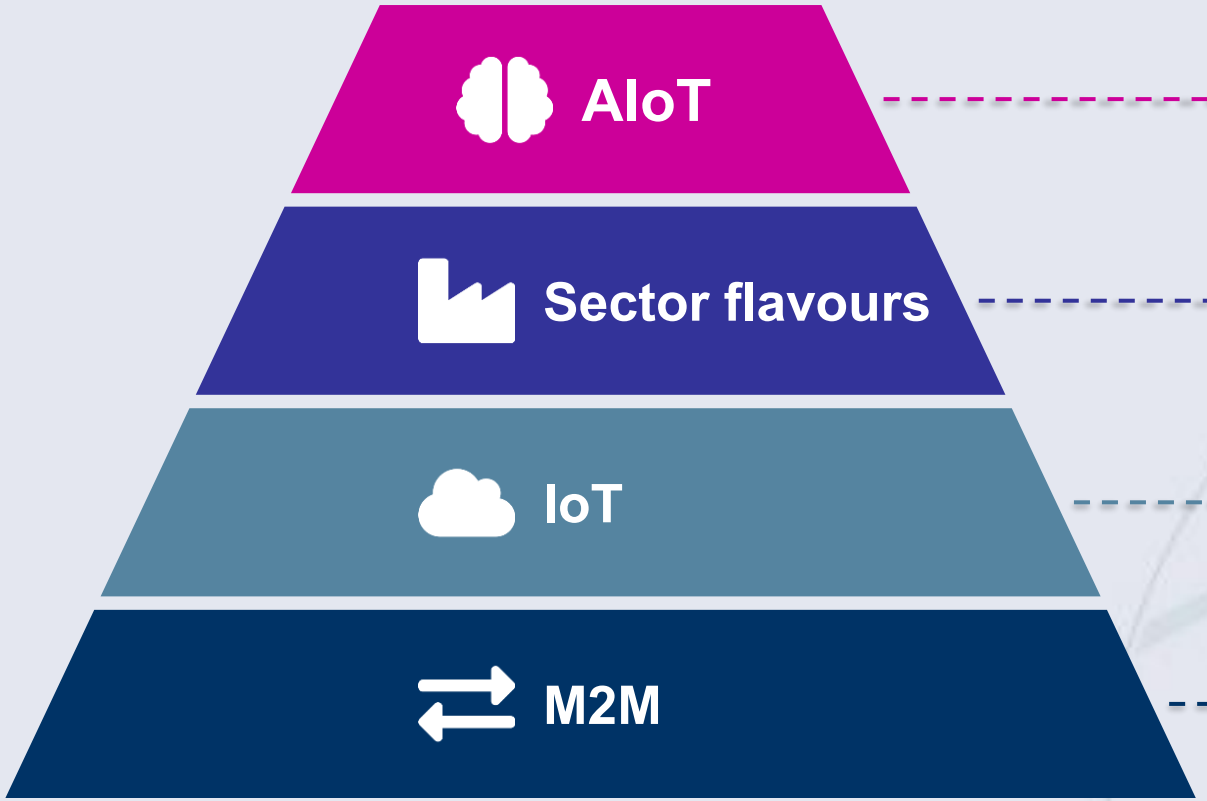
W3C web standards that describe things, so any app can use them.

Today: Siemens and Microsoft link building systems with WoT (Jul 2025)

WoT 1.1 is a W3C standard (Dec 2023); Thing Description 2.0 is still a draft (Nov 2025)

Sources: eCall, europa.eu; IEA Global EV Outlook 2026; W3C WoT; Siemens, Jul 2025; photo credits on the photos

The IoT vocabulary, updated for 2026



things that decide - edge AI and autonomous operations, the 2026 trend in enterprise IoT

IIoT and Industry 4.0; Industry 5.0 (EU: people first); smart city, smart home

things online - devices on IP networks and cloud platforms, the base of this course

machines talk - point-to-point links, mostly cellular; oneM2M is still updated

WoT: the web glue

W3C standards that describe things

new charter 2026-2028

Fading labels: Internet of Everything (Cisco), Industrial Internet

Sources: IoT Analytics 2026; EC Industry 5.0; OMG 2024; W3C; oneM2M

What is IoT? Four recent definitions

USA: NIST Apr 2026	"IoT products are digital equipment or systems that sense or actuate on the physical world while being connected or connectable to the Internet."
EU: Cyber Resilience Act 2024	A "software or hardware product and its remote data processing solutions" that connects, directly or indirectly, to a device or network
Israel: National Cyber Directorate Jun 2025	Embedded-computing products with network connectivity that collect and exchange data with central systems and other devices, locally or in the cloud
Market count: IoT Analytics Oct 2025	Counts connected devices (or the gateway, not each sensor behind it); leaves out computers, phones, tablets, RFID and NFC: 21.1 billion at end-2025

Common core: a physical thing + a sensor or actuator + a network + data

Sources: NIST IR 8259r1, Apr 2026; EU Regulation 2024/2847; INCD, Jun 2025; IoT Analytics, Oct 2025

IoT? (ISO/IEC 20924:2024)

An infrastructure linking things, people, systems and information resources, plus services that process information from the physical and virtual worlds - and react to it

International vocabulary standard, 3rd edition (Feb 2024) - in plain words

Two things to notice

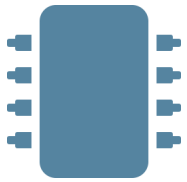
- People are inside the definition
- 'React' = closing the loop, from sensing to action

Other standards bodies

- IEEE: nothing newer than IEEE 2413 (2019)
- ITU-T: Y.4000 (2012) is still in force

Sources: ISO/IEC 20924:2024; IEEE 2413-2019; ITU-T Y.4000

Parts of an IoT product (NIST, FCC)



IoT device

a sensor or actuator plus at least one network interface



Gateway or hub

networking hardware the device works with



Backend

cloud services that store or process the data



Companion app

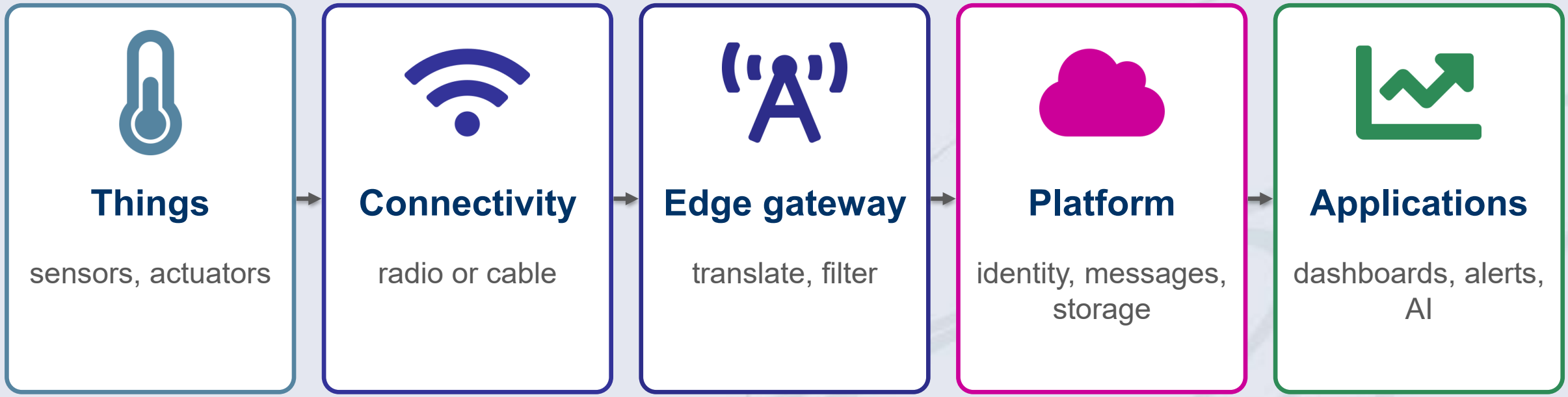
for example the phone app that controls the device

The US security label covers the whole product, not just the device (FCC, 2024)

Next: our five building blocks add the network between the parts - and security across all of them

Sources: NIST IR 8259r1, Apr 2026; FCC 24-26, 2024; ISO/IEC 30141:2024

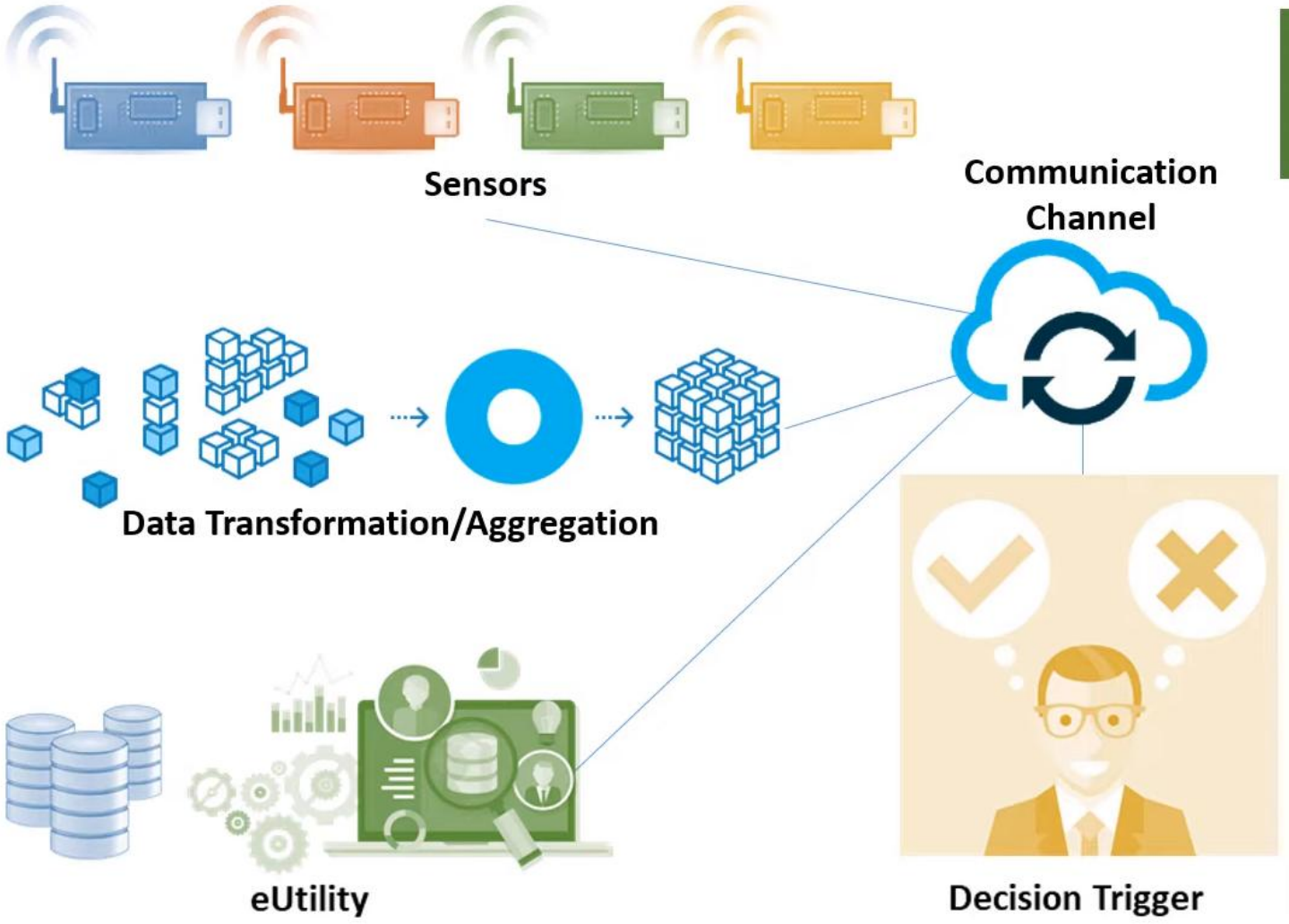
The building blocks of every IoT system



Security and device management run through every block

Our site: door sensor -> LoRa -> mast gateway -> cloud broker -> dashboard + SMS alert

The 5 parts of IoT



Five Knowledge Areas of IoT

Sensors
Aggregators
Communication Channels
eUtility
Decision Trigger

CWNP Official: The *Internet of Things (IoT)* is the application of connected objects (cyber-physical systems and other objects) that use Internet technologies for automation, reporting, and human interaction.

Sensors sense, actuators act

LOGTEL
It's more than knowledge

Sensors: world -> data

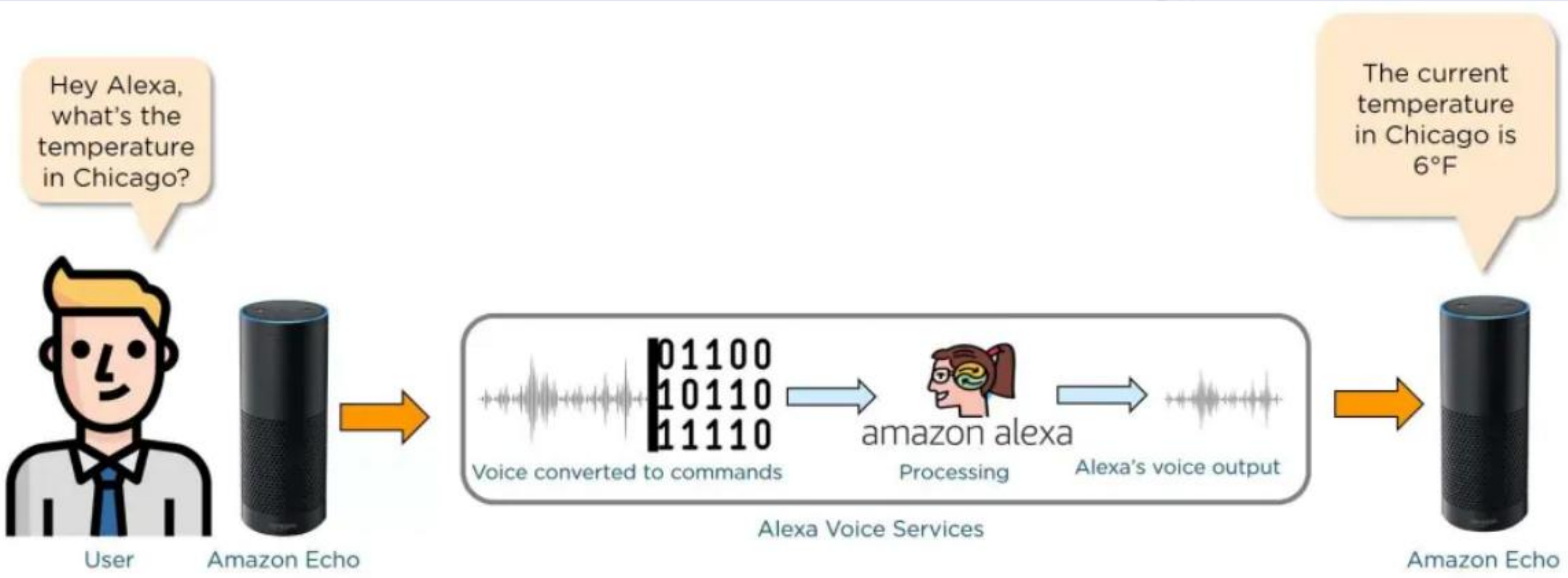
- Temperature
- Motion
- Location
- Light, sound
- Level, flow
- Camera

Actuators: data -> world

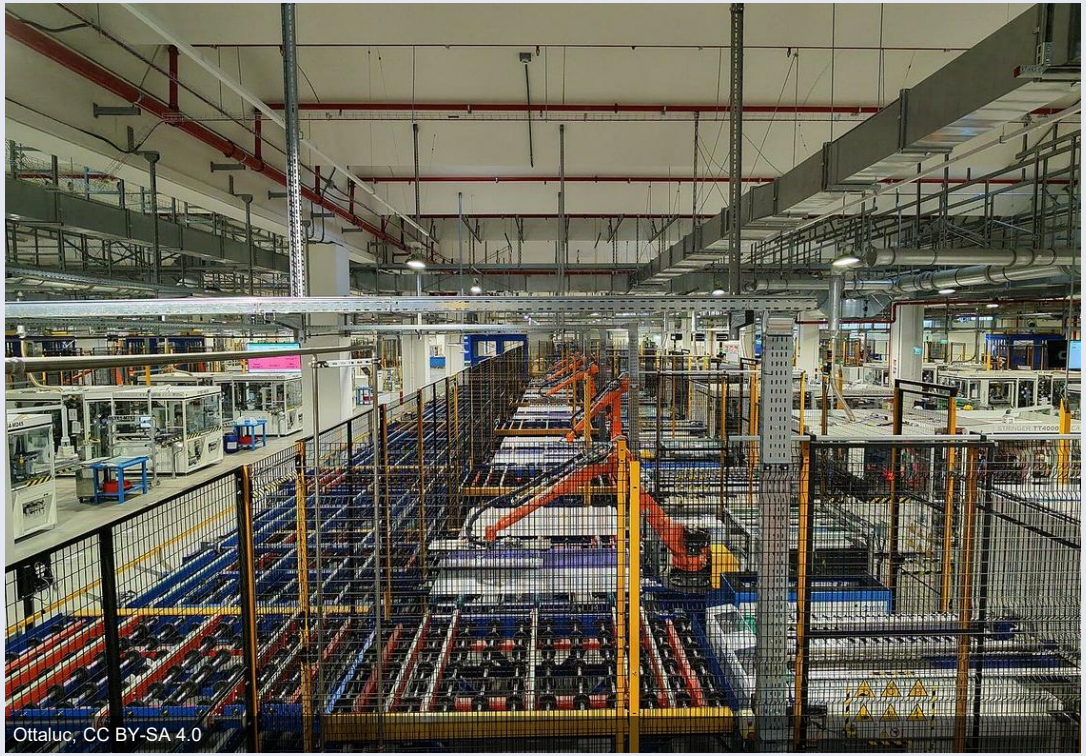
- Lights, displays
- Relays
- Motors, valves
- Sirens
- Locks
- Heating

The loop: sense -> send -> decide -> act.
Decide in the cloud, on the gateway, or on the device (edge AI).

Alexa


Hey Alexa, what's the temperature in Chicago?
User
Amazon Echo
01100
10110
11110
Voice converted to commands
amazon alexa
Processing
Alexa's voice output
Alexa Voice Services
The current temperature in Chicago is 6°F
Amazon Echo

Industrial IoT in 2026: three numbers



5.08 million industrial robots at work worldwide, up 9% in a year (IFR, Sep 2026)

2,031 organisations run their own private 4G/5G network; manufacturing leads (GSA, Sep 2026)

1,019 companies in the OPC Foundation: OPC UA is the shared language of factory machines (Dec 2025)

The EU's next step, Industry 5.0: people first, sustainable, resilient

Sources: IFR 2026; GSA; OPC Foundation; EC Industry 5.0; photo credit on the photo

Smart city: one platform, many users

In the street: sense

- water meters
- waste bins
- street lights
- cameras
- parking
- air quality

connect

City platform

Many users: act

- council
- utilities
- police
- health
- cleaning
- residents



HelgeRieder, CC0

Tel Aviv-Yafo, May 2025

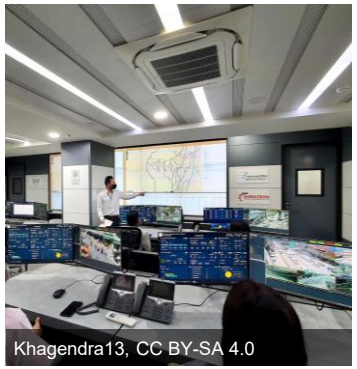
all ~250,000 water meters are smart; leaks found within 48 hours



Tdorante10, CC BY-SA 4.0

Israel, Dec 2025

50-60 local authorities monitor waste with sensors


Khagendra13, CC BY-SA 4.0

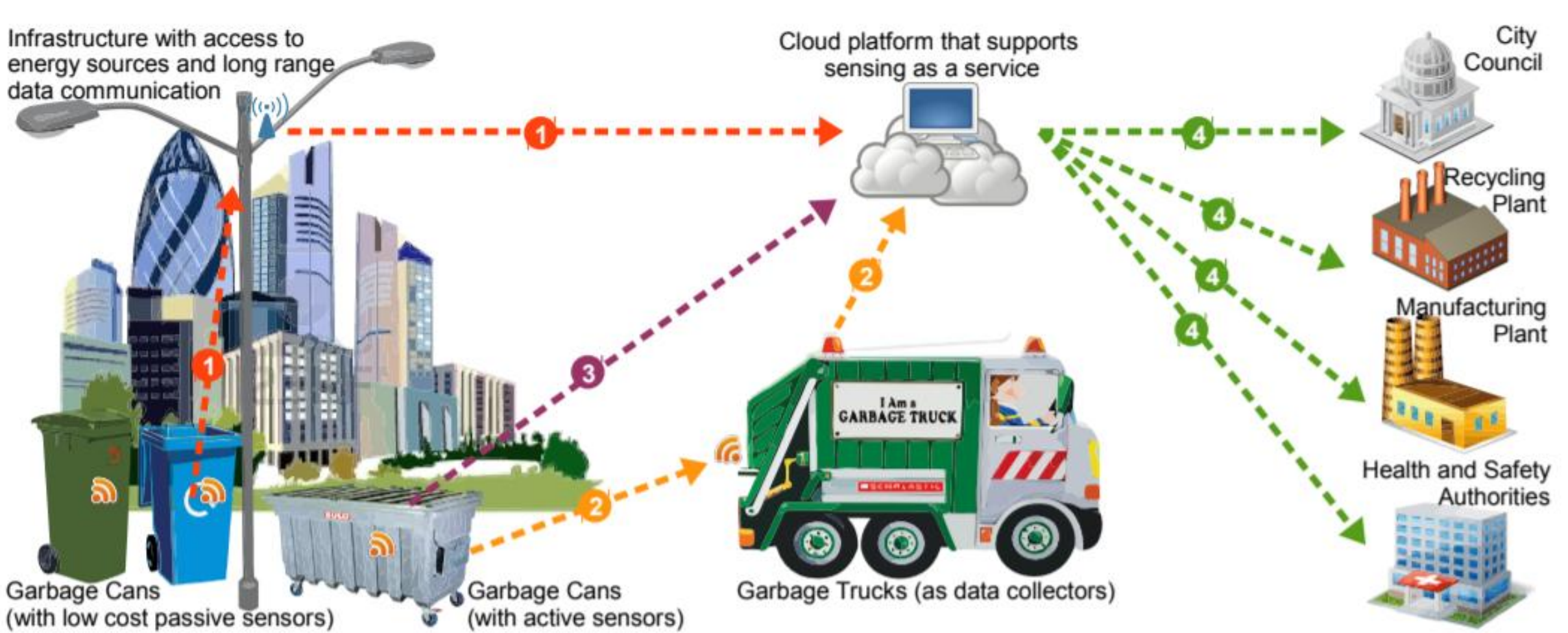
India, Dec 2024

a command and control centre in each of 100 smart cities; 83,000+ CCTV cameras

Pipes, poles and cameras feed one platform - every department uses the same data

Sources: ice.co.il, May 2025; jpost.com, Dec 2025; India PIB, Dec 2024; photo credits on the photos (click a photo)

Smart Cities



Source: Sensing as a Service - Charith Perera , 2018

עמודי תאורה יכולים לעשות כל כך הרבה...

107 אלף עמודי תאורה יוכלו למדוד מהירות – בינתיים המידע לא יעבור למשטרה

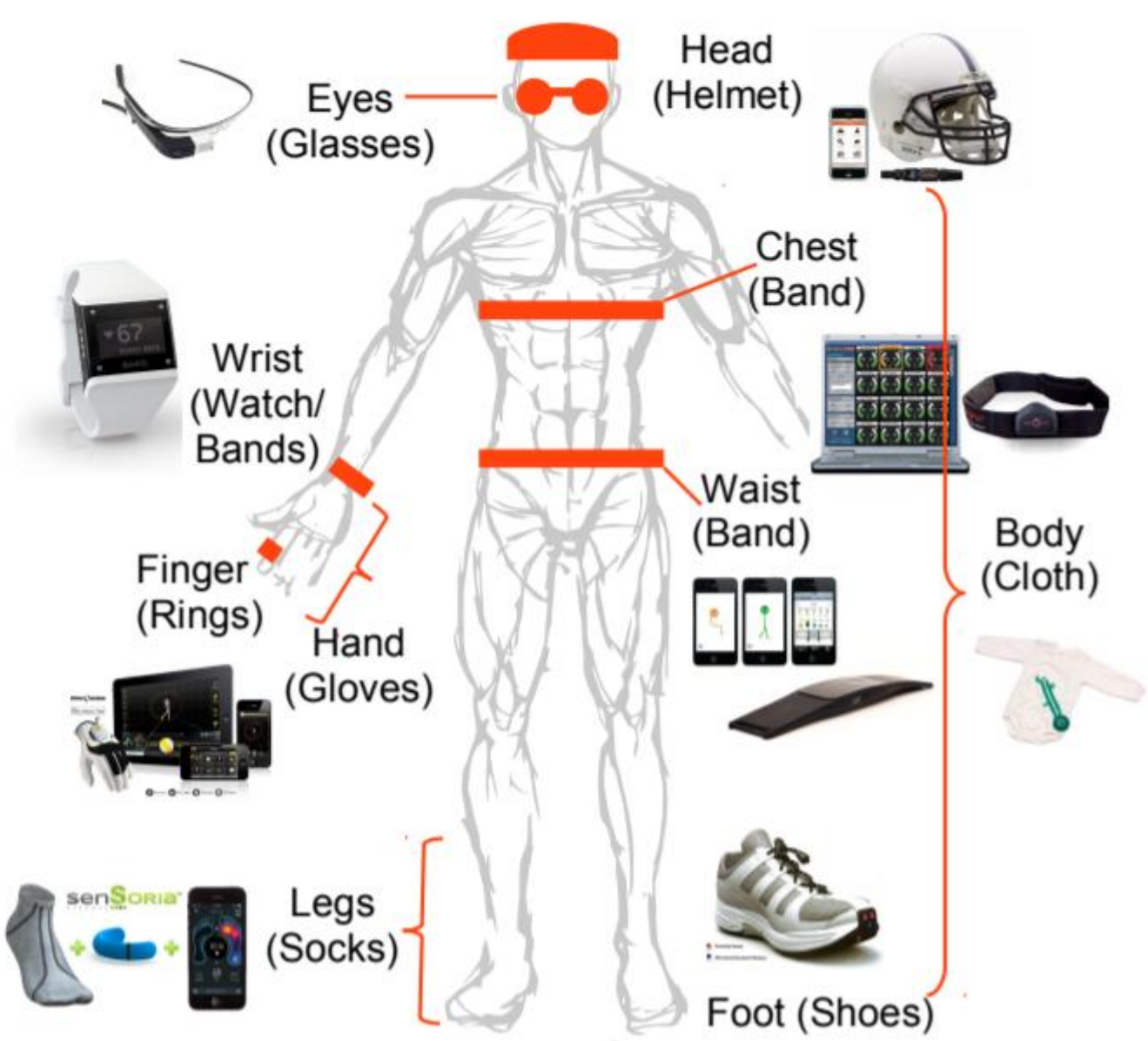
לכלכליסט נודע כי פיילוט למדידת מהירות של כלי רכב כבר החל בכביש 531, בין הרצליה לכביש חוצה ישראל

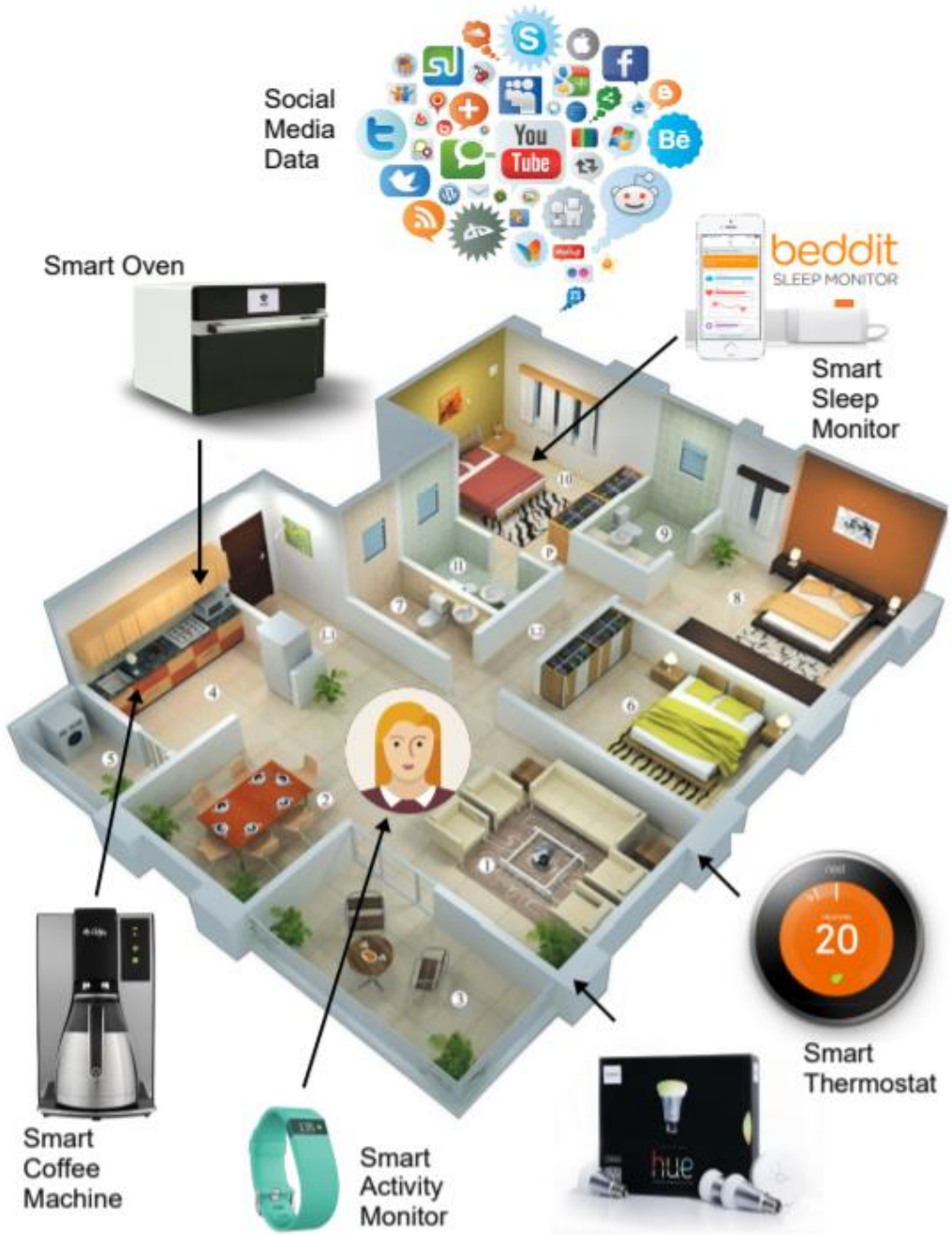
תומר הדר 14:07, 16.01.22

תגיות: מכוניות | מהירות | נתיבי ישראל | עמוד תאורה

לנתיבי ישראל יש 107 אלף עמודי תאורה שיכולים לשמש בעתיד גם למדידת מהירותן של מכוניות. ל"כלכליסט" נודע כי פיילוט מדידת מהירות של כלי רכב כבר החל בכביש 531, שמשתרע בין הרצליה לכביש חוצה ישראל. ועם זאת, יש להדגיש כי המידע על אודות מהירותן של המכוניות לא יימסר לידי משטרת ישראל.

Smart Wearable /Smart Home





Wearables and smart home in 2026

Earbuds

408 M



Smartwatches

160 M

shipped in 2026
(IDC forecast)

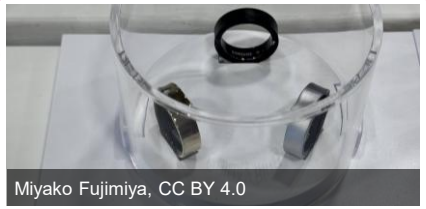


Smart glasses

13.6 M (+41%)

Wristbands

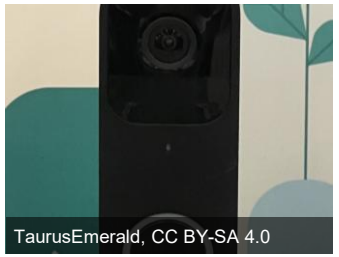
40 M



Smart rings

4.9 M (+13%)

In the home



doorbell



speaker



thermostat

~890 million smart-home devices a year (IDC, 2024)
cameras, locks, doorbells: over a quarter of them

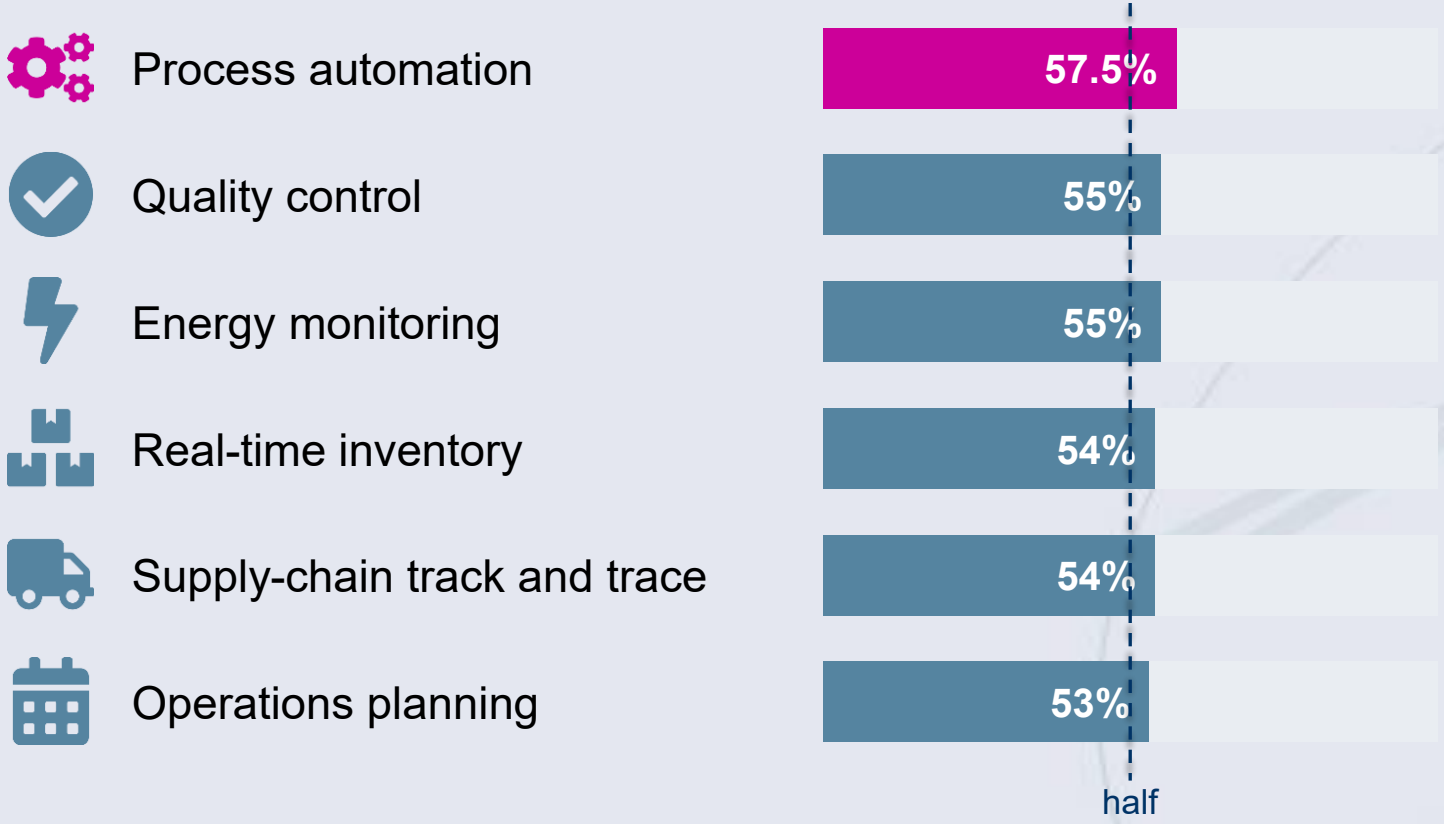
The risk, Mar 2026: one sailor's public Strava run showed where France's aircraft carrier was (Le Monde)

Every wearable is a location sensor - check the privacy settings

Sources: IDC wearables, Jul 2026; IDC smart home, Oct 2024; Stars and Stripes, Mar 2026; photo credits on the photos (click a photo)

The most adopted IoT use cases

Share of IoT adopters using each use case



Survey of 200 companies that already use IoT (IoT Analytics, Jul 2024 - the newest edition)

27 use cases, 3 groups

- Smart operations (14)
- Smart supply chain (3)
- Connected products (10)
- ~92% of adopters see a return

Biggest spenders (EMEA, 2025)

- Retail, utilities, government
- Video analytics: ~8% of IoT spend

Sources: IoT Analytics, Jul 2024; IDC, Mar 2025

IoT at work: six real examples (2025-26)

LOGTEL
It's more than knowledge

Utilities

Great Britain: 42 million smart meters - 72% of all meters

Agriculture

John Deere See & Spray: cameras spray only weeds - about 50% less herbicide

Logistics

Maersk: 4G/5G solar trackers on ~30% of its refrigerated containers

Industry

BlueScope steel: ~2,000 hours of downtime avoided by predictive maintenance

Healthcare

NHS England: ~12,600 'virtual ward' beds - patients monitored at home

Smart city

Prague: fill-level sensors cut glass-bin collections by 65%

IoT in defense and public safety



Ukraine 'Sky Fortress'

~9,500 phone-based acoustic sensors track drones - each under USD 500



Sensor fusion (NATO, 2025)

Exercise in Estonia fused Sky Fortress with radars into one picture



Border sensors

Poland and the Nordics buy mesh-networked ground sensors



Drones = mobile IoT

Sensor + radio + actuator; Ukraine built ~2.2 million in 2024



Predictive logistics

US Army: vehicle sensors forecast failures and spare parts



Networks that survive

Mesh radios keep links alive under jamming

Israel: IoT is already in the street

~70%

of Israel's water meters are smart (goal: 100% in 2026)

1 million+

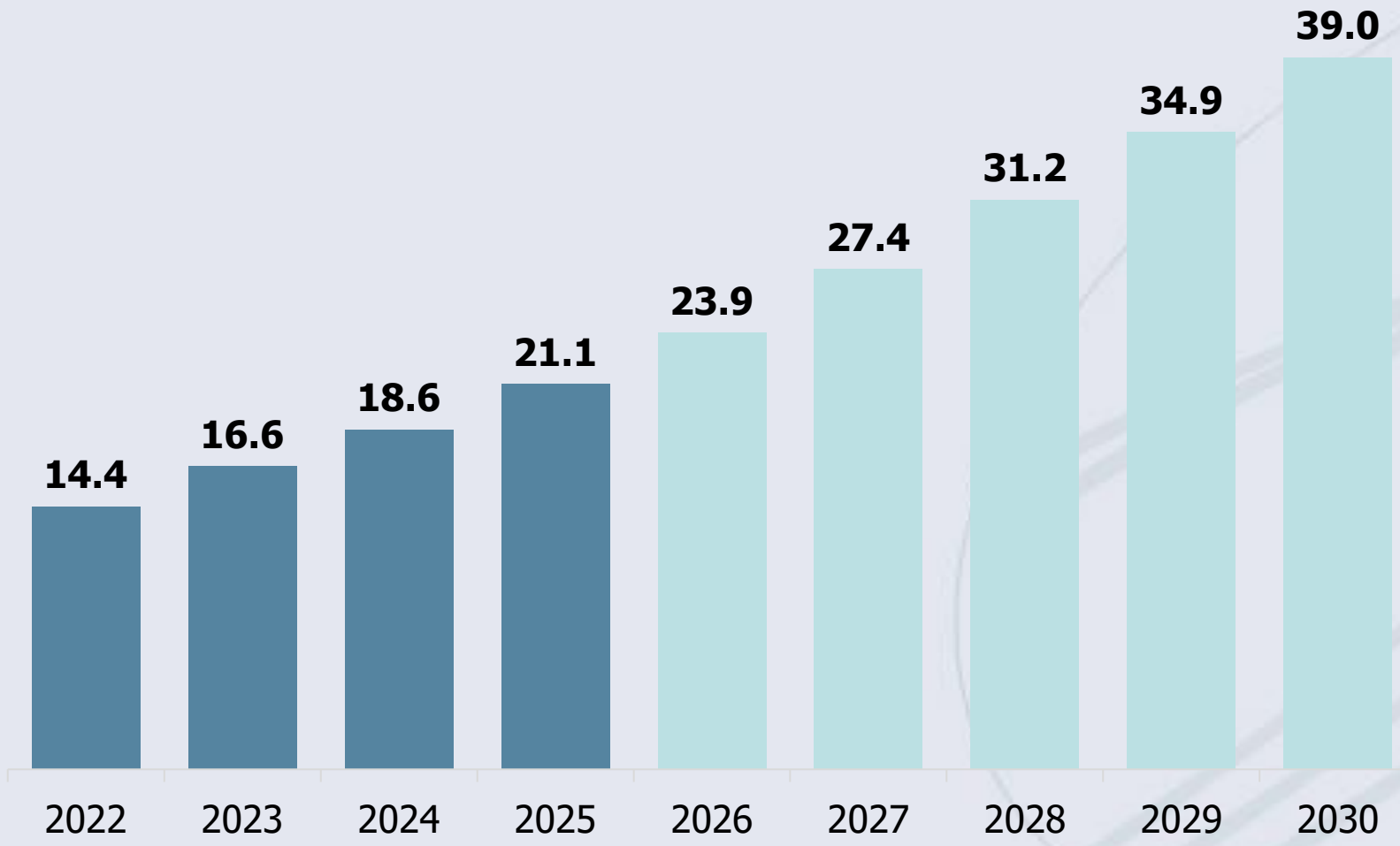
LoRa water meters and sensors in dozens of cities (Nov 2025)

2028

target: smart electricity meters in every home

- Smart water meters flagged leaks in 200,000+ homes in 2024
- The same LoRa networks also serve parking, street lighting, air quality
- Tomorrow: how such a city-wide LoRaWAN network actually works

Connected IoT devices worldwide (billions)



21.1 B

end of 2025

~2.6

devices per person

39 B

forecast for 2030

Source: IoT Analytics, State of IoT 2025 (Oct 2025). Light bars = forecast.

IoT by the numbers

32%

of IoT connections are Wi-Fi

24%

are Bluetooth

22%

are cellular (2G-5G, NB-IoT, LTE-M)

$324 B

enterprise IoT market, 2025 (+13%)

4.5 B

cellular IoT connections, end-2025

< 1%

of IoT devices have real on-device AI

Differences Between H2H and M2M

LOGTEL
It's more than knowledge

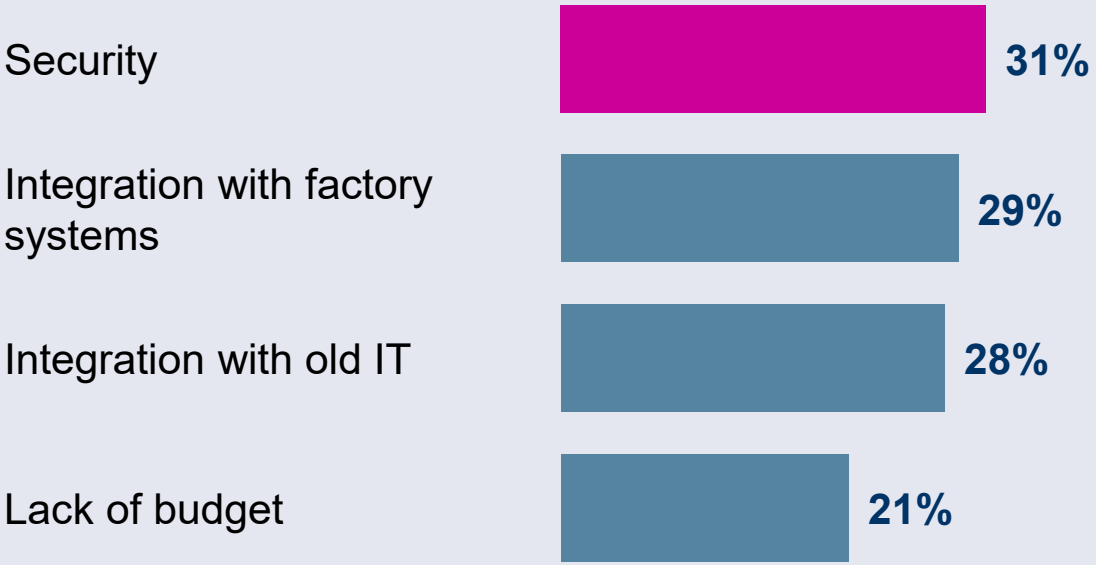
TOPICS	H2H	M2M / IoT
Entity #	There are a lot of us. But maybe not that much compared to M2M potential	M2M outnumbers human end users by order of magnitude
Data Volume	Most of the traffic is downloaded and requires significant amounts of bandwidth (file, Web and video streaming).	Traffic is mainly uploaded and most of it requires small amounts of bandwidth (video surveillance might require more).
Battery	We can buy a new one. We can recharge.	Auto-generating power or be self-sustaining for long periods.
Delay	We do have a tolerance for it even when it comes to voice	some applications are mainly for real-time control; urgent/emergency action would have little tolerance.
security	Worth comes to worth. Instinct tells us or we just know if it has been stolen.	Not so much here. Machine Communication Identity Module (MCIM) in Universal Integrated Circuit Card (UICC) or Trusted Environment (TRE).
Revenue	Good.	Low ARPD.
Value Chain	Well defined.	To be created/adapted, new opportunities on top of common infrastructure.
Reachability	satisfying.	Push/pull behavior. Might require longer dormant period, minimize signaling on control plane.

Characteristics (IoT v/s Cellular)

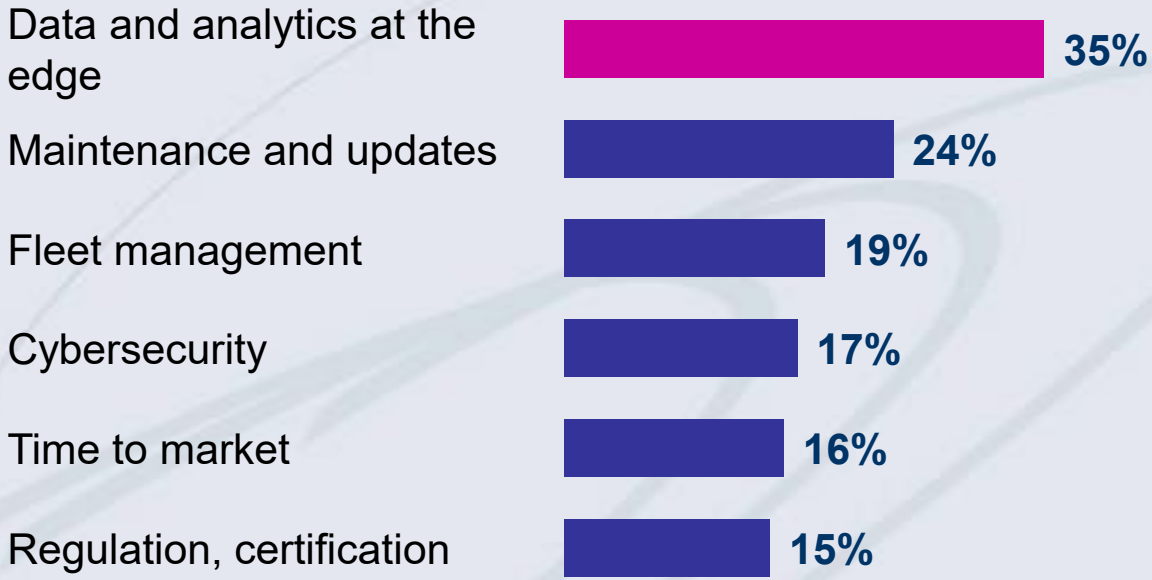
- **IoT communications are or should be:**
  - Low cost,
  - Low power,
  - Long battery duration,
  - High number of connections,
  - Different bitrate requirement,
  - Long range,
  - Low processing capacity,
  - Low storage capacity,
  - Small size devices,
  - Simple network architecture and protocols

What holds IoT back (surveys 2025-26)

Buyers - Omdia, Oct 2026 (edge IoT users)



Builders - Eclipse developer survey, Mar 2026



GSMA Intelligence, 4,200 enterprises: cost of implementation comes first (39%); security and integration follow, about 1 in 3 each; lack of in-house skills is 4th

Running and integrating IoT is harder than buying sensors.

Sources: Omdia via IoT Business News, Oct 2026; Eclipse Foundation 2025 survey; GSMA via IoT Now, 2025

When IoT becomes a weapon (2024-2026)

31.4 Tbps

record DDoS (Q4 2025) - a botnet of mostly Android TV boxes

130 -> 805

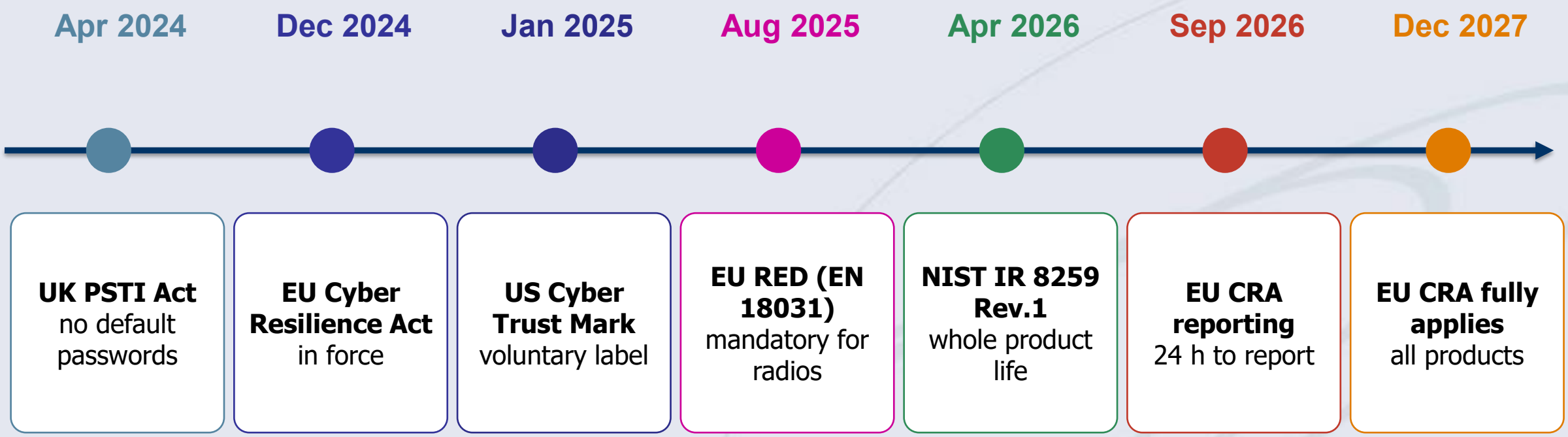
attacks above 1 Tbps Q1 -> Q2 2026

millions

devices sold pre-infected (FBI, BADBOX 2.0, 2025)

Poland, Dec 2025	30+ wind and solar sites attacked; controllers damaged, no blackout
Norway, Apr 2025	hackers opened a dam valve for about 4 hours via a remote panel
USA, Sep 2024	FBI takedown of a 260,000-device camera and router botnet

Regulation has arrived: IoT security timeline



What they all ask for:

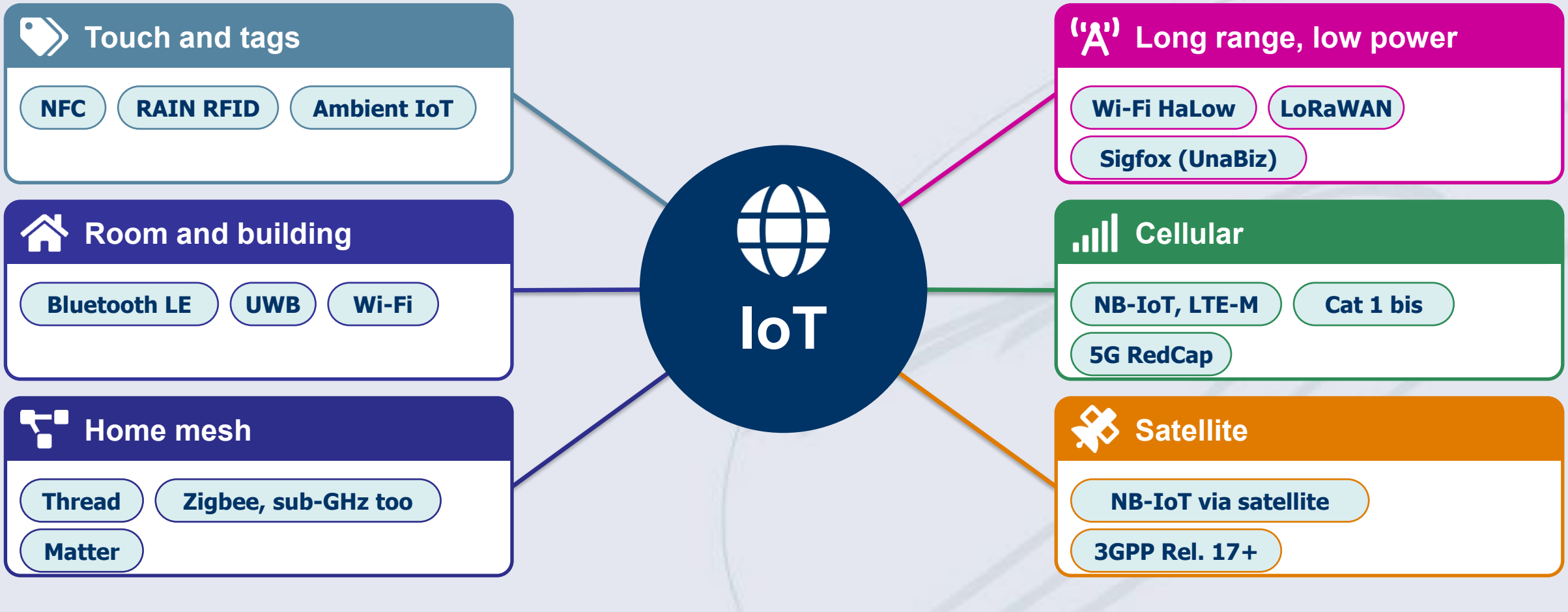
- **no default passwords**
- **a way to report flaws**
- **security updates for a stated period**
- **secure by design**

Who writes the IoT standards?

Architecture and vocabulary	ISO/IEC JTC 1/SC 41 (ISO/IEC 30141:2024), ITU-T (Y.4000)
Applications and messaging	OASIS (MQTT), CSA (Matter, Zigbee), OMA (LwM2M), Eclipse (Sparkplug)
Internet protocols	IETF: IPv6, 6LoWPAN, CoAP, TLS
Radios and cellular	IEEE (Wi-Fi, 802.15.4), Bluetooth SIG, LoRa Alliance, 3GPP (NB-IoT, LTE-M, 5G)
Security baselines	ETSI EN 303 645, NIST IR 8259, IEC 62443, EN 18031

Many bodies, many standards - the price of a huge, diverse market.

IoT technologies: the 2026 map



Speeds and ranges: next slides. Long range in depth: Day 2. Cellular: Day 3 with Gilad.

Sources: 3gpp.org; csa-iot.org; wi-fi.org; telecomtv.com (Sigfox)

Short-range technologies at a glance

	Typical range	Data rate	Power	Typical IoT use
NFC	a few cm	up to 848 kbit/s	tag needs none	payment, access, pairing
RFID (UHF)	cm to several m	1,000 tags/s	tag needs none	inventory, logistics
Bluetooth LE	10-100 m	0.125-2 Mbit/s	very low	wearables, trackers, sensors
Zigbee / Thread	10-100 m + mesh	250 kbit/s	very low	home and building automation
Wi-Fi	20-150 m	up to Gbit/s	high	cameras, gateways
Wi-Fi HaLow	~1 km	up to tens of Mbit/s	low-medium	campus, farm, cameras
UWB	up to 100 m	up to ~27 Mbit/s	low	cm-accurate location

Typical values - real range depends on antennas, walls and settings.

Range vs speed: the 2026 map

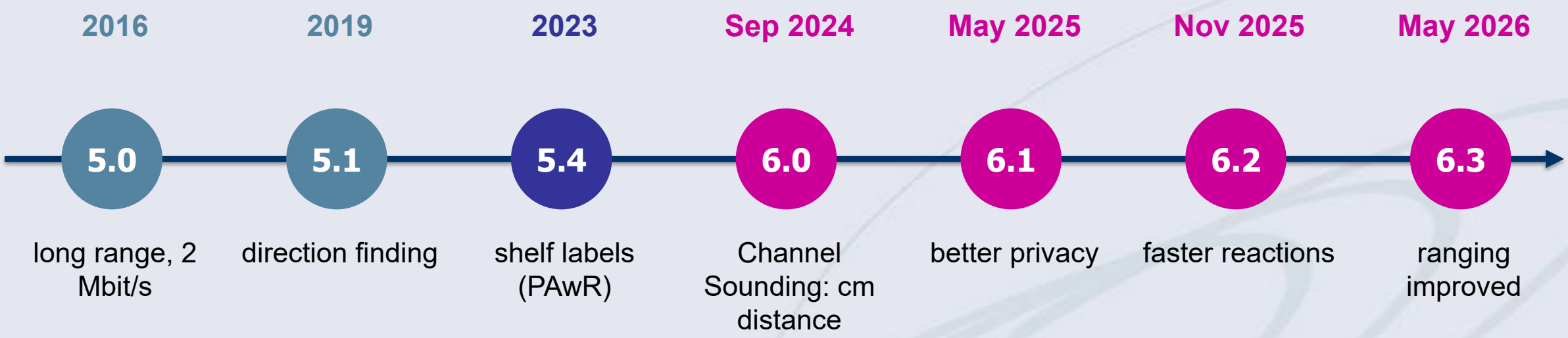
	Touch (cm)	Room, building (10-100 m)	Site (about 1 km)	Wide area (km)	Anywhere (satellite)
Gbit/s		Wi-Fi 6/7			
Mbit/s		UWB	Wi-Fi HaLow	5G RedCap	
0.1-1 Mbit/s	NFC	Bluetooth LE		LTE-M	
under 100 kbit/s				NB-IoT LoRaWAN	NB-IoT

Wide area and satellite: typical speed. Short range: top speed - real use is lower.

Israel (Ministry of Communications, Dec 2025): no hard 2G/3G switch-off; the 800/900 MHz bands move to 5G from Feb 2026; a 2G network for emergency calls runs to the end of 2028

Sources: iot.telenor.com; lora-alliance.org; 3gpp.org; ynet.co.il, Dec 2025

Bluetooth keeps evolving



5.4 B

Bluetooth devices shipped in 2025

8 B+

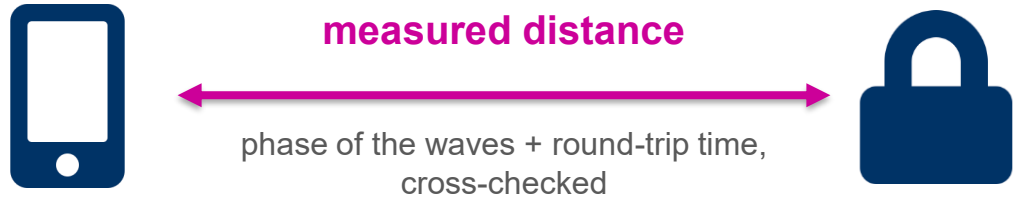
per year expected by 2030

Auracast

public broadcast audio is spreading

Bluetooth 6.0 and 6.2 up close

6.0 - Sep 2024: Channel Sounding, true distance

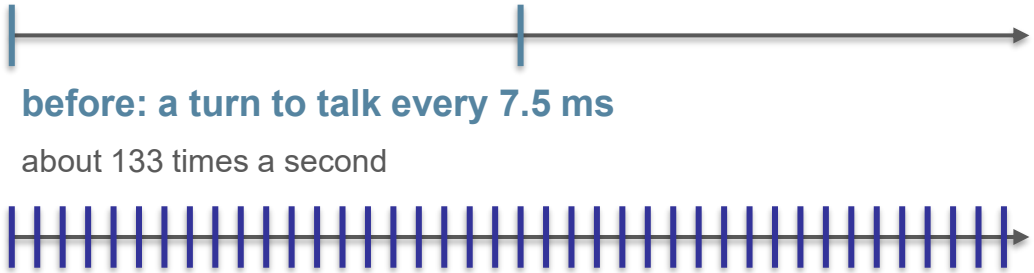


Before: distance guessed from signal strength (RSSI) - metres off, easy to fake

Now: typically within 20-50 cm, and a relay attack shows up



6.2 - Nov 2025: 20x shorter connection intervals





1.2-3.0 2003-09 classic: headsets, cars	**4.0** 2010 Low Energy is born	**4.2** 2014 secure pairing	**5.0** 2016 2 Mbit/s or 4x range	**5.1** 2019 direction finding	**5.2** Dec 2019 the base of LE Audio
5.3 2021 less power: subrating	**5.4** 2023 e-shelf labels (PAwR)	**6.1** May 2025 more private addresses	**6.3** May 2026 faster ranging	**next** late 2026 HDT: up to 7.5 Mbit/s	6.0 and 6.2: detailed above

Sources: SIG Core 6.0 overview, Jan 2025; SIG Channel Sounding, Oct 2024; SIG Core 6.2, Nov 2025; SIG on shorter intervals, May 2026; SIG HDT, Jul 2026

Bluetooth LE: advertise, connect, read

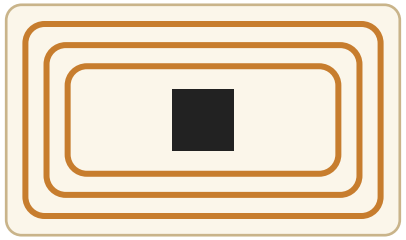


Data on the device

- Temperature service
  - temperature = 23
- Accelerometer service
  - x, y, z
- UART service
  - text in / out

- **Advertising** = shouting your name; **connection** = a private, hopping conversation
- Very low power: the radio sleeps most of the time; range about 10-100 m

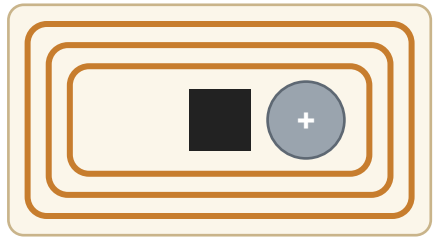
Passive, semi-passive or active?



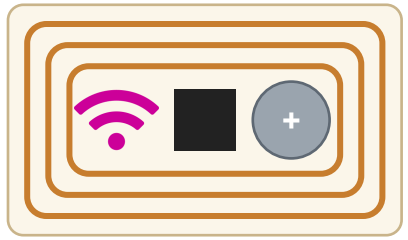
antenna + chip



Polarbear24, CC BY-SA 4.0



antenna + chip + battery



antenna + chip + battery + radio



Kyu3a, CC BY-SA 4.0

Passive

- No battery, no transmitter
- The reader's signal powers the tag; the tag answers by reflecting it
- NFC, RAIN RFID, 3GPP ambient IoT 'device 1' (about 1 microwatt)

Semi-passive

- Still reflects instead of transmitting
- A small amplifier adds reach
- 3GPP Release 20 target ('device 2a')

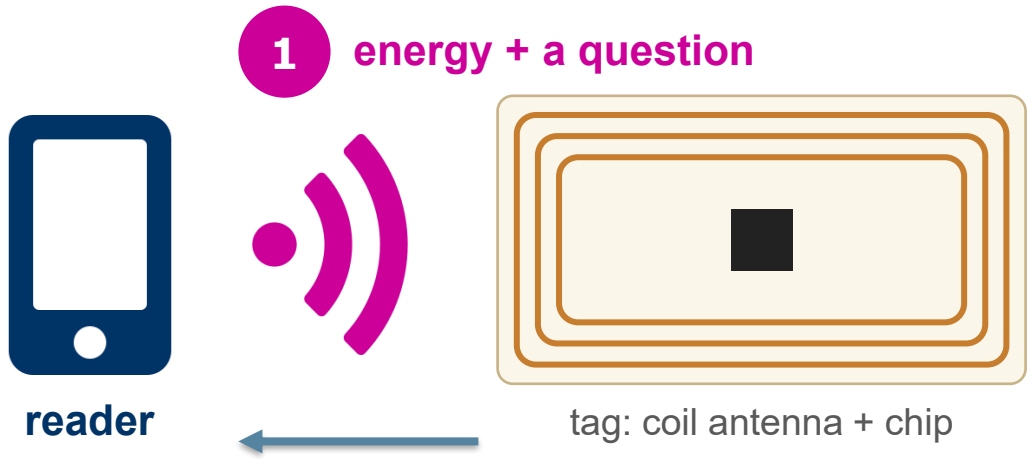
Active

- Own battery and radio
- Bluetooth LE, LoRa, cellular sensors
- One BLE chip sleeps at 200 nA (vendor figure)

Less energy on the device = shorter reach and simpler messages

Sources: 3gpp.org Release 19 ambient IoT; nfc-forum.org; bluetooth.com; photo credits on the photos

RFID and NFC: powered by the reader



2 the chip wakes up on that energy - no battery

3 it answers by changing how it loads or reflects the signal

NFC

- 13.56 MHz, touch range (about 2 cm)
- The phone is the reader



4300streetcar, CC BY 4.0

RAIN RFID (UHF)

- 860-930 MHz, by country; metres
- Up to 1,000 tags per second

One reader can read thousands of tags at once - without seeing them

Sources: nfc-forum.org; therainalliance.org

NFC, RFID and battery-free 'ambient' IoT

0.5 -> 2 cm

NFC Release 15 (June 2025)

43-53 B

RAIN RFID chips shipped a year (2024-2025)

< 10 cents

battery-free Bluetooth tag

- **Ambient IoT**: tags with no battery, powered by surrounding radio waves
- Walmart (Oct 2025): millions of battery-free sensors in its supply chain
- Being standardized now: 3GPP Releases 19-20 (cellular) and IEEE 802.11bp (Wi-Fi, ~2028)
- NFC roadmap (Feb 2026): up to 8x faster data and post-quantum security

UWB: centimetre-accurate location



- Very short pulses (6-8 GHz); the receiver measures their flight time
- Distance and direction to about 10 cm - far better than signal strength
- Car keys and door locks open only when the phone is really next to them

Aliro 1.0 - Feb 2026	FiRa Core 4.0 - Dec 2025	AirTag 2 - Jan 2026
one phone credential for door locks: NFC, BLE and UWB	the current UWB spec; adds asset tracking	second-generation UWB chip, 1.5x finding range

Competition: Bluetooth Channel Sounding (improved again in 6.3, May 2026) also measures distance - typically within 20-50 cm, cheaper, no extra radio

Wi-Fi and BLE in Wireshark: same idea

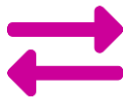
	Wi-Fi capture	Bluetooth LE capture
1	radio info (channel, signal)	radio info (channel, RSSI)
2	802.11 frame (MAC)	Link Layer
3	IP	-
4	TCP / UDP	L2CAP, ATT
5-7	TLS, MQTT, DNS ...	GATT values

BLE carries no IP - a gateway is needed to reach the Internet.

Day 1 in six sentences



Layers let us swap radios without rewriting the application.



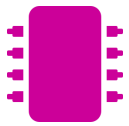
Devices dial out and keep a session alive.



TCP buys reliability; UDP buys speed and battery.



TLS gives privacy, integrity and identity.



Sensors reach the controller via GPIO, UART, I2C, SPI.



Every radio trades range, data rate and power.

Quick quiz

1. The device reaches the cloud, but the cloud cannot reach the device. Why?
2. Five sensors, two wires, each with an address. Which interface?
3. Which Wireshark filter shows only name look-ups?
4. Why does 868 MHz usually reach farther than 2.4 GHz?
5. What three things does TLS give you?
6. Which layer chooses between TCP and UDP?

Thank You!!!